Toward the Alaoglu–Erdős Integer-Exponent Conjecture

Corpus Audit, Strong Algebraic-Base Spectra, Tower Rigidity, and a Branch-Sensitive Research Program

A detailed analysis of

https://github.com/VladimirReshetnikov/ProveIt

at repository commit

`4848c6d834190301e014028a9c78a3f0ef49b7b8`

Verdict. I did not obtain an unconditional proof of the conjecture. The obstruction is not a missing elementary lemma: a hypothetical counterexample supplies precisely the unresolved 2×2 logarithmic determinant underlying the four exponentials conjecture. The repository has nevertheless moved well beyond that bare reduction. Its Lean development formalizes a sharp conditional semigroup structure, finite exclusions, arithmetic rigidity, counting and equidistribution statements, radical reformulations, a generic algebraic six-exponentials determinant, and the exact positive-algebraic output locus.

At the audited snapshot, Lean already proves that under a nonintegral solution the positive algebraic bases with algebraic x-th power are exactly $\Gamma = \{2^r 3^s : r, s \in \mathbb{Q}\}$. This report gives a short general two-base derivation of that phenomenon and adds results not isolated in the current article corpus: Roy's strong six-exponentials upgrade; a Smith-normal-form lattice for rational and integral outputs; a solution-independent second-iterate kernel; its rational Möbius-orbit characterization; depth-three power-tower rigidity; and new one-base tower equivalences. Except where marked **[Lean]**, these additions are complete paper proofs, not yet machine-checked or peer-reviewed.

Prepared for Vladimir Reshetnikov

OpenAI GPT-5.6 Pro

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Abstract

The Alaoglu–Erdős integer-exponent conjecture, originating in the 1944 work of Alaoglu and Erdős [1], asserts that a real number x for which 2^x and 3^x are integers must itself be an integer. I audited all three LaTeX reports under `Analysis/ExponentialIdentities/Article/**/` and the current Lean import surface at the commit printed on the title page. The corpus proves, among other things, that a counterexample is transcendental; that failure produces a unique least nonintegral solution β and the exact solution monoid $\mathbb{N}_0 + \mathbb{N}_0\beta$; that the corresponding candidate integers form a free rank-two monoid; that nonintegral candidates have zero density and obey strong progression and determinant repulsion estimates; and that the algebraic-output locus among all positive algebraic auxiliary bases is exactly the positive $2, 3$-unit divisible hull, with additional mixed-second-iterate rank restrictions. The Lean kernel excludes nonintegral solutions with $0 < x < 12$, while a separate directed-rounding MPFR computation reported in the third article excludes nonintegral solutions with $0 < x \le 26$; the latter is source-audited here but not reclassified as a formal proof.

A central theorem of the current Lean snapshot is the exact $2, 3$ algebraic-base spectrum: under a nonintegral solution, a positive algebraic α has algebraic α^x exactly when some positive power of α is a rational $2, 3$-unit, equivalently when $\alpha \in \Gamma = \{2^r 3^s : r, s \in \mathbb{Q}\}$. I give a short general two-base version. If a, b are multiplicatively independent positive algebraic numbers, x is irrational, and a^x, b^x are algebraic, then

$$\alpha^x \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \alpha \in \Gamma_{\mathrm{sat}}(a, b) := \{a^r b^s : r, s \in \mathbb{Q}\}.$$

The difficult direction is an exact application of six exponentials and is compatible with the repository's generic algebraic determinant. The genuinely new upgrade uses Baker's homogeneous theorem and Roy's strong six exponentials theorem to prove $x \log \alpha \notin \widetilde{\mathcal{L}}$ outside the saturated group, where $\widetilde{\mathcal{L}}$ is the $\overline{\mathbb{Q}}$-linear span of 1 and logarithms of algebraic numbers. Consequently every fixed algebraic $\alpha \notin \Gamma$ is a complete one-number detector for the conjecture, generalizing the repository's base-5 equivalence.

For a hypothetical nonintegral solution, put $M = 2^x$ and $A = 3^x$. Inside Γ, rational and integral outputs admit the exact valuation criteria

$$(2^r 3^s)^x = M^r A^s \in \mathbb{Q} \Longleftrightarrow r\,\nu(M) + s\,\nu(A) \in \mathbb{Z}^{(\mathbb{P})},$$

with nonnegativity added for integrality. This gives a finite-index rank-two superlattice of $\mathbb{Z}^2$, whose quotient is read off from the Smith invariants of the prime-valuation matrix. A second, coarser kernel

$$K_x = \{(r, s) \in \mathbb{Q}^2 : M^r A^s \in \Gamma\}$$

is independent of the chosen nonintegral solution. It has dimension at most one, contains no vector with $rs > 0$, and is nonzero exactly when

$$x = \frac{u + v\vartheta}{r + s\vartheta}, \qquad \vartheta = \log_2 3,$$

for rational coefficients with nonzero determinant. Thus $K_x \neq 0$ is equivalent to $x \in \mathrm{PGL}_2(\mathbb{Q}) \cdot \vartheta$ and to a rational linear dependence among $1, \vartheta, x, x\vartheta$. The external prime-valuation vectors of M and A then give a four-way classification: a $\{2, 3\}$-smooth M; a $\{2, 3\}$-smooth A; nonsmooth outputs with a common external valuation ray; or two independent external valuation directions.

Finally, if $\alpha \in \overline{\mathbb{Q}}_{>0} \setminus \{1\}$, then at least one of $\alpha^x, \alpha^{x^2}, \alpha^{x^3}$ is transcendental, and the logarithm of the first transcendental term lies outside $\widetilde{\mathcal{L}}$. The exact failure depth is 1, 2, or 3 according as $\alpha \notin \Gamma$, $\alpha \in \Gamma \setminus K_x$, or $\alpha \in K_x \setminus \{1\}$. This yields, for example, the new equivalence that Alaoglu–Erdős holds if and only if every two-base solution makes both 2^{x^2} and 2^{x^3} algebraic. The report closes with a branch-sensitive research program, emphasizing a two-prime tropical/p-adic determinant in the generic branch and pure-radical or common-core methods in the exceptional branches.

Contents

1 Executive answer and trust boundaries

1.1 What was and was not proved

The requested primary goal was an unconditional proof. After reconstructing the present formal and paper-level frontier, trying direct logarithmic, determinant, radical, valuation, and orbit arguments, and then pushing the auxiliary-base and iterated-output structures, I do not have such a proof. I found no valid step that turns the exact relation

$$\log M\ \log 3 = \log A\ \log 2, \qquad M = 2^x \in \mathbb{N}, \quad A = 3^x \in \mathbb{N},$$

into a forbidden linear form in logarithms. This relation is bilinear, and its normalized 2×2 logarithm matrix is precisely the unresolved four-exponentials configuration. The open status of the general four exponentials conjecture was rechecked against a reviewed July 2026 problem record and the standard transcendence literature [17, 10, 11].

That negative verdict should not obscure the amount of genuine progress. The report contains three kinds of statements:

- **[Lean]** statements already checked by the Lean kernel in the repository;
- **[corpus paper proof]** statements proved in the three inspected LaTeX reports but not necessarily formalized;
- **[new paper proof]** statements proved in this report and independently checked at the paper level, but not yet machine-checked or peer-reviewed.

The computational exclusion through 2^{26} is marked **[interval computation]**. It uses directed MPFR intervals and a complete finite scan according to the supplied source, but it is not a Lean proof and I did not rerun all 67,108,863 nonzero inputs in this environment. I did audit the rounding logic, endpoint logic, and nearest-integer coverage described by the source.

1.2 The strongest additions in this report

The following are the main additions relative to the inspected article corpus.

Result	Content
Saturated-group synthesis	The current Lean divisible-hull theorem is recast as the $2, 3$ case of a general two-base spectrum $\Gamma_{\text{sat}}(a, b) = \{a^r b^s : r, s \in \mathbb{Q}\}$, with a short six-exponential proof.
Strong spectrum	Outside Γ, not only is α^x transcendental, but $x \log \alpha \notin \widetilde{\mathcal{L}}$ by Baker plus Roy.
Valuation lattice	Within Γ, rational and integral outputs are classified exactly by the prime-valuation matrix of (M, A); Smith normal form gives the finite rational-output defect.
Symmetric constants	For $\vartheta = \log_2 3$, at least one of 3^ϑ and $2^{1/\vartheta}$ has logarithm outside $\widetilde{\mathcal{L}}$; under a counterexample, each is algebraic exactly in one explicit smooth-output branch.

Result	Content
Second-iterate kernel	The set of 2^r3^s whose x-th powers return to Γ is a solution-independent vector space of dimension at most one.
Möbius reformulation	The kernel is nonzero exactly when $x \in \mathrm{PGL}_2(\mathbb{Q}) \cdot \vartheta$, equivalently when $1, \vartheta, x, x\vartheta$ are rationally dependent.
Depth-three tower theorem	Every nontrivial positive algebraic base has a transcendental term among $\alpha^x, \alpha^{x^2}, \alpha^{x^3}$; the first failing depth is classified exactly.
New conjecture equivalences	Any algebraic base outside Γ is a one-output detector; for bases inside Γ, three tower levels are enough. In particular, base 2 alone at levels x^2 and x^3 detects the conjecture.

The exact $2, 3$ algebraic-output locus itself is **[Lean]** in the current snapshot; this report's new content begins with its general saturated-group packaging, the Roy-strengthened form, and the subsequent valuation-kernel consequences. Here "new" means not isolated in the inspected article corpus and derived during this analysis. No claim of priority in the broader literature is intended.

1.3 Repository snapshot

The audit used commit

`4848c6d834190301e014028a9c78a3f0ef49b7b8`,

whose root toolchain selects Lean 4.32.0. The article glob contains three source reports:

1. `alaoglu-erdos-combined-report/alaoglu-erdos-combined-report.tex`;
2. `alaoglu-erdos-further-report/alaoglu-erdos-further-report.tex`;
3. `alaoglu-erdos-report-3/alaoglu-erdos-report-3.tex`.

They are cumulative rather than independent [13, 14, 15]: the combined report consolidates the original formal program and, at this snapshot, includes the generic positive-real algebraic determinant, the exact arbitrary positive-algebraic output locus, and second-iterate rank constraints; the further report develops arbitrary-node and mixed-exponent interpolation determinants; the third report pushes nonlinear closure, primitive-solution statistics, finite computation, and mixed-output transcendence.

2 The problem and its exact transcendence-theoretic shape

2.1 Elementary reductions

Let

$$\mathcal{S} = \{x \in \mathbb{R} : 2^x \in \mathbb{Z},\ 3^x \in \mathbb{Z}\}.$$

Because both powers are positive, their integral values are positive. If $x < 0$, then $0 < 2^x < 1$, so $x \notin \mathcal{S}$. Hence $\mathcal{S} \subseteq [0, \infty)$.

Lemma 2.1 (Rational solutions are integral)**.** *If $x \in \mathbb{Q}$ and $2^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$. Consequently every nonintegral element of $\mathcal{S}$ is irrational.*

Proof. Write $x = p/q$ in lowest terms with $q > 0$. If $2^{p/q} = M \in \mathbb{N}$, then $2^p = M^q$. Unique factorization gives $q \mid p$, hence $q = 1$. $\square$

If x were algebraic irrational, the Gelfond–Schneider theorem [3, 5] would make 2^x transcendental. Therefore:

Proposition 2.2 (Transcendence of a counterexample)**.** *Every nonintegral $x \in \mathcal{S}$ is transcendental.*

This is both elementary from Gelfond–Schneider and formalized in the repository in the appropriate real-power language.

2.2 Candidate integers and the slope ϑ

Set

$$\vartheta = \frac{\log 3}{\log 2} = \log_2 3.$$

Then ϑ is irrational by unique factorization and transcendental by Gelfond–Schneider [3], since $2^\vartheta = 3$. If $M = 2^x$, then

$$3^x = M^\vartheta.$$

Thus the original problem is equivalent to classifying the candidate set

$$\mathcal{C} = \{M \in \mathbb{N}_{>0} : M^\vartheta \in \mathbb{N}\}.$$

The conjecture is

$$\mathcal{C} = \{2^n : n \in \mathbb{N}_0\}.$$

The logarithmic relation for a candidate-output pair (M, A) is

$$\frac{\log M}{\log 2} = \frac{\log A}{\log 3} = x.$$

Equivalently,

$$\det \begin{pmatrix} \log 2 & \log 3 \\ \log M & \log A \end{pmatrix} = 0. \tag{2.1}$$

Under a nonintegral solution, both rows and both columns are linearly independent over $\mathbb{Q}$. All four entries are logarithms of algebraic numbers. The four exponentials conjecture predicts that such a singular matrix cannot exist [10, 11]. This is the exact global wall.

2.3 Why integrality still matters

Although (2.1) identifies an open transcendence core, the hypotheses are much stronger than arbitrary algebraicity: M and A are positive integers, and every $n + kx$ is again a solution. This supplies:

- a discrete additive semigroup of exponents;
- two synchronized multiplicative semigroups of integer outputs;
- prime valuations and perfect-power descent;
- exact denominator control after reducing modulo integer translations;

- interpolation determinants with integral entries;
- a dense irrational rotation after taking fractional parts.

The repository's main achievement is to exploit nearly all of this structure without silently assuming the missing four-exponentials step.

3 Audit of the current ProveIt frontier

3.1 Kernel-verified backbone

The aggregate Lean file imports a broad collection of modules under `TwoBaseIntegerExponent`. The most important groups are as follows.

Theme	Representative formal content
Finite exclusion	Exact checks below 256, 4096, and related localized intervals; in particular a nonintegral solution has $x \geq 12$.
Rationality and transcendence	Rational solutions are integral; nonintegral solutions and $\log 3/\log 2$ are transcendental.
Localization and radicals	Odd-core reduction, canonical radical form, radical degree, 3-denominator normalization, and valuation sharpness.
Progressions and density	Sharp progression obstructions, higher differences, zero density, noninteger density, shrinking-target and denominator-growth estimates.
Conditional structure	Least solution, primitive generator, exact counting, quadratic asymptotics, block Weyl statements, output normalization, common-power rigidity.
Auxiliary outputs	Rational and algebraic third bases, generic positive-real algebraic six-exponentials, exact rational- and algebraic-base output loci, and rank-three reformulations.
Second iterates and external cores	Mixed-output half-plane transcendence, rank-one ratio loci and an explicit external-prime valuation line, core independence, and canonical-radical divisible-hull exclusions.

Three current formal interfaces are especially important for this report. First, for every positive algebraic real γ under the two integrality hypotheses,

$$\gamma^x \in \overline{\mathbb{Q}} \iff x \in \mathbb{Z} \text{ or } \quad \exists n \geq 1,\ \gamma^n \in \langle 2, 3\rangle_{\mathbb{Z}}.$$

For a nonintegral solution, the right-hand exceptional locus is exactly $\Gamma = \{2^r 3^s : r, s \in \mathbb{Q}\}$. This is formalized through `HasPositiveTwoThreeUnitPower` and `TwoBaseNonintegerSolution.real_rpow_isAlgebraic_iff_hasPositiveTwoThreeUnitPower`. The natural-base classification

$$a^x \in \overline{\mathbb{Q}} \iff x \in \mathbb{Z} \text{ or } a = 2^u 3^v \quad (u, v \in \mathbb{N}_0)$$

is a special case.

Second, the new `AlgebraicOutputLocus` module proves that every genuinely mixed base $(2^i 3^j)^{x^2}$ with $i, j > 0$ is transcendental under a nonintegral solution, and that two algebraic members of the ratio family $(2^i/3^j)^{x^2}$ must have collinear exponent vectors. Third, the corresponding equivalences include the base-5 algebraicity formulation, the base-6 squared-exponent formulation, the positive-algebraic rank-three locus, and the rank-two ratio-locus formulation of Alaoglu–Erdős.

3.2 Exact conditional semigroup structure

The central conditional theorem can be summarized as follows [13].

Theorem 3.1 (Conditional rank-two structure; corpus synthesis)**.** *Assume the conjecture fails. Then there is a unique least nonintegral solution $\beta > 0$, and*

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\}.$$

Writing

$$M = 2^\beta, \qquad A = 3^\beta,$$

one has exact output formulae

$$2^{n+k\beta} = 2^n M^k, \qquad 3^{n+k\beta} = 3^n A^k,$$

and therefore

$$\mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\}.$$

The representation is unique.

The existence of a least solution uses discreteness forced by the candidate integers; closure under subtraction of ordered solutions then supplies Euclidean descent. Uniqueness of the (n, k) representation uses the transcendence of β. This is considerably more rigid than a generic hypothetical set of exceptional exponents.

The least pair (M, A) is affinely primitive: one cannot simultaneously remove a common integer translation from the exponent or a common perfect-power factor from both outputs. The current arithmetic-rigidity module formalizes a quantitative version: if

$$2^x = 2^r a^k, \qquad 3^x = 3^r b^k,$$

then $(x - r)/k$ is another solution; the verified zero-free interval yields

$$r + 8k \le x$$

for a nonintegral solution. Later paper-level finite work improves the raw exclusion but the formal affine theorem deliberately depends only on the kernel-verified bound available in that module.

3.3 Counting, spacing, and interpolation determinants

Conditional on failure, exact rank-two structure gives logarithmic local counts and quadratic cumulative counts. The further report also proves unconditional candidate repulsion from arbitrary-node generalized Vandermonde determinants [14]. A typical statement is that if

$$N \le m_0 < m_1 < \cdots < m_r \le N + H, \qquad m_j \in \mathcal{C},$$

then

$$\prod_{0\le i<j\le r}(m_j-m_i)\ge \frac{r!}{|\vartheta(\vartheta-1)\cdots(\vartheta-r+1)|}m_0^{r-\vartheta}.$$

Using mixed exponents $a+b\vartheta$ strengthens the local spacing scale; for sufficiently large r the report obtains a bound of the schematic form

$$H\ge \frac{1}{24}N^{1-12/\sqrt{r}}.$$

The resulting unconditional dyadic count is $O((\log X)^2)$, while a counterexample's exact monoid predicts only $O(\log X)$ points in $[X,2X]$. This single missing logarithm is a useful diagnosis: the determinant method is close enough to see the correct geometry, but not close enough to contradict it.

3.4 Nonlinear closure and the third report

The third report [15] exploits the transcendence of β inside the exact monoid. Among its cleanest paper-level statements are:

$$(n+k\beta)(m+\ell\beta)\in\mathcal{S}\quad\Longleftrightarrow\quad k=0 \text{ or } \ell=0,$$

so two nonintegral solutions never multiply to another solution. More generally, polynomial and rational-function intersections with $\mathcal{S}$ collapse to affine functions in β. It also derives primitive-solution asymptotics, exact divisor-lattice formulae, root-closure criteria, optimized high-difference exclusions, and mixed-exponent transcendence such as the transcendence of 6^{xy} for nonintegral solutions x, y.

Some of these statements have since acquired neighboring Lean infrastructure, but the whole third-report package is not yet represented by one formal module. This report therefore keeps its status labels theorem-by-theorem rather than treating the PDF as a machine-checked object.

4 Attempts at a complete proof and the precise points of failure

This section records the principal proof attempts. Several produce useful lemmas later in the report; none produces the missing contradiction. It is important to separate a route that is merely unfinished from one that has silently restated the open conjecture.

4.1 Direct linear forms in logarithms

The relation

$$\log M\,\log 3-\log A\,\log 2=0$$

is not a linear form in logarithms. Baker's theorem controls expressions

$$c_1\log\alpha_1+\cdots+c_n\log\alpha_n$$

with algebraic coefficients c_i; here the coefficients $\log 2$ and $\log 3$ are themselves transcendental logarithms. Dividing by $\log 2\log 3$ merely returns

$$\frac{\log M}{\log 2}=\frac{\log A}{\log 3}.$$

Any argument saying that the equality of these logarithmic ratios forces multiplicative dependence of $(2, M)$ or $(3, A)$ is exactly a four-exponentials assertion and is not presently available.

The integrality of M, A does permit prime expansions

$$\log M = \sum_p v_p(M) \log p, \qquad \log A = \sum_p v_p(A) \log p,$$

but substitution yields a quadratic relation among logarithms:

$$\sum_p v_p(M) \log p \log 3 - \sum_p v_p(A) \log p \log 2 = 0.$$

Known homogeneous linear-independence results do not control such a relation. The support information will become useful only after an additional algebraic output is manufactured.

4.2 Trying to manufacture a third algebraic output

The six exponentials theorem [7, 5, 9] immediately finishes the problem if one can find a positive algebraic base α, multiplicatively independent of 2 and 3, such that α^x is algebraic. Indeed, using rows $\log 2, \log 3, \log \alpha$ and columns $1, x$ would give six algebraic exponentials, contradicting six exponentials.

The current Lean module `AlgebraicThirdBase.lean` realizes this principle internally for natural bases by an interpolation determinant and upgrades rationality to integrality. Its consequence is not a proof of Alaoglu–Erdős; it says that a counterexample forces *every* independent natural auxiliary output to be transcendental.

Natural candidates for a third base include $M + A$, $M - A$ when positive, prime divisors of M or A, and primitive odd cores. They are algebraic bases, but no identity makes their x-th powers algebraic. The exact spectrum theorem in the next section shows that this is not merely a failure of imagination: under a counterexample, no algebraic base outside the saturated group generated by 2 and 3 can have algebraic output. Thus a successful "third-base" proof must derive algebraicity from an additional arithmetic operation, not just choose a clever base.

4.3 Radical normalization and local ramification

A candidate M can be written in a canonical radical form. Removing the largest power of 2, extracting the maximal perfect power from the odd core, and normalizing powers of 3 reduces the conjecture to a statement of the form

$$u^\vartheta \notin \mathbb{Z}[1/3] \qquad \text{for every odd power-primitive } u > 1$$

subject to explicit side conditions. The repository proves exact radical-degree and valuation statements around this reduction.

A tempting route is to place a hypothetical value

$$u^\vartheta = \frac{a}{3^t}$$

in a pure extension and compare ramification at 3 and at primes dividing u. This does not work directly because ϑ is transcendental: the notation is an equality of positive real numbers, not an algebraic radical extension generated by a rational exponent. One can obtain genuine radicals after a smooth-output or common-perfect-power reduction, but then the remaining relation is again a product of logarithms. Local valuations rigorously control every *algebraic* power that appears; they do not assign a p-adic meaning to the real exponent ϑ itself.

A productive reformulation is therefore branch-sensitive. In the smooth-output branches of section 7, one of the constants 3^{ϑ} or $2^{1/\vartheta}$ becomes a genuine algebraic radical, and local Kummer theory can legitimately be brought to bear. In the generic branch, two independent external valuation directions are available instead.

4.4 Integer-gap estimates from rational approximation

Let p/q approximate x. Then

$$M^q - 2^p \in \mathbb{Z}, \qquad A^q - 3^p \in \mathbb{Z}.$$

If neither difference vanishes, each has absolute value at least 1. Taylor expansion gives bounds of the rough form

$$|qx - p| \gg 2^{-p}, \qquad |qx - p| \gg 3^{-p}.$$

These are valid denominator-growth estimates but exponentially weaker than the polynomial lower bounds that would conflict with continued fractions. The denominator of the rational approximation is q, whereas the integer scale is exponential in q.

One can cancel several Taylor terms with Hermite–Padé or interpolation determinants. The further report does exactly this in a coordinate-free way and obtains strong local repulsion, but the exact counterexample monoid remains sparse enough to fit within the bounds. A cluster built from a convergent p/q to the least solution β has baseline size exponential in q; its small relative width does not become a sufficiently small *absolute* integer quantity.

4.5 Archimedean interpolation determinants

For candidate pairs (m_i, n_i) with $n_i = m_i^{\vartheta} \in \mathbb{N}$, matrices with entries

$$m_i^{a_j} n_i^{b_j} = m_i^{a_j + b_j \vartheta}$$

have integer determinants. If the nodes m_i are close, confluent Vandermonde estimates make the determinant small in the ordinary absolute value. Nonvanishing then gives a lower bound of 1, producing spacing inequalities.

There are three recurring limitations.

1. The number of useful mixed exponents grows quadratically with the exponent box, while the analytic losses also grow with determinant size.
2. A generic dyadic block estimate sees all possible candidate configurations, not the exact triangular (n, k) lattice forced by a counterexample.
3. The archimedean norm alone discards prime-valuation information in the entries.

The first two explain the one-logarithm deficit between the unconditional $O((\log X)^2)$ dyadic bound and the conditional $O(\log X)$ count. The third motivates the proposed adelic/tropical determinant in section 13.

4.6 p-adic exponentiation and why it cannot simply replace the real exponent

For a p-adic unit close to 1, the function $z \mapsto u^z$ is p-adically analytic on a suitable domain. However, the real number x has not been shown to define a compatible p-adic exponent, and an integer such as $M = 2^x$ need not determine such a compatibility. Writing the real relation inside a p-adic logarithm without first producing an algebraic identity is invalid.

What *is* legitimate is to apply p-adic valuations to the integer interpolation matrices, to algebraic radicals already certified by rational exponents, or to multiplicative relations such as $M^r A^s = 2^u 3^v$. The second-iterate kernel introduced later cleanly detects when the latter relations occur.

4.7 Additive and multiplicative closure

The solution set is additively closed, while the third report proves that, under failure, the product of two nonintegral solutions is never a solution. One might hope to derive an independent closure under squaring or multiplication from the original power identities. No such identity is available:

$$2^{x^2} = (2^x)^x = M^x$$

raises an integer base to the same unknown exponent and is generally transcendental. The repository's theorem that 6^{x^2} is transcendental shows that nonlinear closure fails in a particularly robust way. The tower theorem in section 9 explains the exact depth at which every algebraic base must fail.

4.8 Bottom line of the proof attempts

A complete proof would follow from any one of the following genuinely new bridges:

- algebraicity of α^x for one algebraic $\alpha \notin \Gamma$;
- a sub-$O(\log X)$ unconditional dyadic count for candidates;
- a local/p-adic obstruction to every one of the four kernel branches in section 8.4;
- a proof that one of 2^{x^2} and 2^{x^3} is algebraic under the original hypotheses in a way that contradicts the depth-three theorem;
- the relevant special case of four exponentials, strong four exponentials, or a sufficiently strong algebraic-independence conjecture for logarithms.

The rest of this report turns the failed third-base and iterated-output attempts into exact classification theorems.

5 Exact algebraic auxiliary-base spectrum

5.1 The saturated algebraic group

The multiplicative group $\overline{\mathbb{Q}}_{>0}$ becomes a vector space over $\mathbb{Q}$ when rational scalar multiplication is interpreted by the unique positive real root:

$$q \cdot \alpha = \alpha^q, \qquad q \in \mathbb{Q}, \quad \alpha \in \overline{\mathbb{Q}}_{>0}.$$

For multiplicatively independent $a, b \in \overline{\mathbb{Q}}_{>0}$, define their saturated group

$$\Gamma_{\mathrm{sat}}(a,b) = \{a^r b^s : r, s \in \mathbb{Q}\}.$$

The map $\mathbb{Q}^2 \to \Gamma_{\mathrm{sat}}(a,b)$ is an isomorphism of $\mathbb{Q}$-vector spaces.

The current Lean snapshot already contains the essential determinant in a fully symmetric form: three positive algebraic bases with injective nonnegative monomials cannot all have algebraic x-th powers for an irrational x. It also packages the specialization to $2, 3$ as an exact divisible-hull classification. The next theorem is the concise coordinate-free synthesis for an arbitrary independent pair a, b; its proof is a direct application of the same classical six-exponentials input.

Theorem 5.1 (Exact algebraic-base spectrum)**.** *Let $a, b \in \overline{\mathbb{Q}}_{>0}$ be multiplicatively independent, let $x \in \mathbb{R} \setminus \mathbb{Q}$, and assume $a^x, b^x \in \overline{\mathbb{Q}}$. Then, for every $\alpha \in \overline{\mathbb{Q}}_{>0}$,*

$$\alpha^x \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \alpha \in \Gamma_{\mathrm{sat}}(a,b).$$

[new paper proof]

Proof. If $\alpha = a^r b^s$ with $r, s \in \mathbb{Q}$, positivity gives

$$\alpha^x = (a^x)^r (b^x)^s,$$

which is algebraic.

Conversely, suppose $\alpha \notin \Gamma_{\mathrm{sat}}(a,b)$. Then

$$\log a, \quad \log b, \quad \log \alpha$$

are linearly independent over $\mathbb{Q}$. Indeed, a rational relation can be cleared to an integer multiplicative relation. If the coefficient of $\log \alpha$ is zero, it contradicts the multiplicative independence of a, b; otherwise it places α in the saturated group. The pair $1, x$ is linearly independent over $\mathbb{Q}$. Apply the six exponentials theorem to the three rows $\log a, \log b, \log \alpha$ and the two columns $1, x$. Five of the six exponential values

$$a,\ b,\ \alpha,\ a^x,\ b^x$$

are algebraic. Therefore α^x is transcendental. □

This theorem is conceptually stronger than a list of auxiliary bases: it says that two algebraic outputs have already exhausted the maximum algebraic multiplicative rank allowed by six exponentials.

5.2 Specialization to Alaoglu–Erdős

For the original problem write

$$\Gamma = \Gamma_{\mathrm{sat}}(2,3) = \{2^r 3^s : r, s \in \mathbb{Q}\}.$$

Corollary 5.2 (Complete algebraic spectrum of a counterexample)**.** *Let $x \in \mathcal{S} \setminus \mathbb{Z}$. For every $\alpha \in \overline{\mathbb{Q}}_{>0}$,*

$$\alpha^x \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \alpha \in \Gamma.$$

In particular, the spectrum is independent of the hypothetical nonintegral solution x. ***[Lean]***

Proof. By theorem 2.1, x is irrational. Applying theorem 5.1 with $a = 2$, $b = 3$ proves the statement directly. Equivalently, the current Lean predicate `HasPositiveTwoThreeUnitPower` says that some positive natural power of α is a positive rational $2, 3$-unit. For positive algebraic α, this is equivalent to $\alpha \in \Gamma$: one implication raises $2^r 3^s$ to a common denominator, and the other uses the unique positive real root. The theorem `TwoBaseNonintegerSolution.real_rpow_isAlgebraic_iff_hasPositiveTwoThreeUnitPower` therefore gives exactly the displayed equivalence. □

The group Γ is countable, divisible, of rational rank two, and dense in $\mathbb{R}_{>0}$; density already follows from the subgroup $\{2^r : r \in \mathbb{Q}\}$. Thus a counterexample would have a dense set of algebraic bases with algebraic output, but every algebraic base outside that exact dense subgroup would have transcendental output.

Corollary 5.3 (Rank bound for any family of algebraic outputs)**.** *Let $x \in \mathcal{S} \setminus \mathbb{Z}$, and let $\alpha_1, \ldots, \alpha_t \in \overline{\mathbb{Q}}_{>0}$ satisfy $\alpha_i^x \in \overline{\mathbb{Q}}$. Then the multiplicative rank of the α_i is at most two, and every α_i lies in Γ.*

5.3 Every independent algebraic base is a detector

Theorem 5.4 (Algebraic detector equivalence)**.** *Fix $\alpha \in \overline{\mathbb{Q}}_{>0} \setminus \Gamma$. Then the Alaoglu–Erdős conjecture is equivalent to*

$$\forall x \in \mathbb{R}, \quad 2^x, 3^x \in \mathbb{Z} \Longrightarrow \alpha^x \in \overline{\mathbb{Q}}.$$

[new paper proof]

Proof. If the conjecture holds, every solution x is a nonnegative integer, so α^x is algebraic. Conversely, a nonintegral solution would make α^x transcendental by theorem 5.2. □

This is a paper-level universalization of the kernel-verified pointwise locus, rather than a new transcendence estimate. The repository's base-5 equivalence is the natural-integer instance $\alpha = 5$. The theorem shows that no arithmetic feature specific to 5 is needed. Valid detectors include, for example,

$$5, \quad \sqrt{5}, \quad 1 + \sqrt{2}, \quad \frac{7 + 3\sqrt{5}}{2},$$

provided the selected number is not accidentally in Γ. For an algebraic integer having a prime divisor outside $2, 3$, exclusion from Γ is immediate; for a general algebraic number it may be checked by multiplicative independence.

Corollary 5.5 (Zero-one law across nonintegral solutions)**.** *Assume the conjecture fails, and let $\alpha \in \overline{\mathbb{Q}}_{>0}$. Exactly one of the following holds:*

1. *$\alpha \in \Gamma$, and α^x is algebraic for every nonintegral solution x;*
2. *$\alpha \notin \Gamma$, and α^x is transcendental for every nonintegral solution x.*

5.4 Strong spectrum via Baker and Roy

Let $\widetilde{\mathcal{L}}$ denote the $\overline{\mathbb{Q}}$-vector space generated by 1 and all logarithms of nonzero algebraic numbers, allowing arbitrary complex branches. For positive real algebraic numbers we use the real logarithm, which is one of those branches.

Roy's strong six exponentials theorem [8, 11] says that if three complex numbers are linearly independent over $\overline{\mathbb{Q}}$ and two more are linearly independent over $\overline{\mathbb{Q}}$, then at least one of the six pairwise products lies outside $\widetilde{\mathcal{L}}$.

Theorem 5.6 (Strong algebraic-base spectrum)**.** *Under the hypotheses of theorem 5.1, if $\alpha \notin \Gamma_{\mathrm{sat}}(a, b)$, then*

$$x \log \alpha \notin \widetilde{\mathcal{L}}.$$

In particular, α^x is transcendental. ***[new paper proof]***

Proof. As in theorem 5.1, the three logarithms $\log a, \log b, \log \alpha$ are $\mathbb{Q}$-linearly independent. Baker's homogeneous theorem [2] upgrades them to linear independence over $\overline{\mathbb{Q}}$.

The number x is transcendental. Indeed, if it were algebraic, its irrationality together with $a \neq 0, 1$ would make a^x transcendental by Gelfond–Schneider. Hence $1, x$ are linearly independent over $\overline{\mathbb{Q}}$.

Apply Roy's theorem to rows $\log a, \log b, \log \alpha$ and columns $1, x$. Five products lie in $\widetilde{\mathcal{L}}$:

$$\log a,\ \log b,\ \log \alpha, \quad x \log a = \log(a^x), \quad x \log b = \log(b^x).$$

Therefore the sixth product $x \log \alpha$ lies outside $\widetilde{\mathcal{L}}$. □

Definition 5.7. For brevity, this report calls a positive exponential value e^z *log-strongly transcendental* when $z \notin \widetilde{\mathcal{L}}$. This implies ordinary transcendence of e^z, because the logarithm of a positive algebraic number belongs to $\widetilde{\mathcal{L}}$.

Corollary 5.8 (Strong detector)**.** *For every fixed $\alpha \in \overline{\mathbb{Q}}_{>0} \setminus \Gamma$, Alaoglu–Erdős is equivalent to the assertion that all two-base solutions satisfy*

$$x \log \alpha \in \widetilde{\mathcal{L}}.$$

5.5 Five-exponential rays forced by a counterexample

The classical five exponentials theorem [11] gives another useful description of the hypothetical world.

Proposition 5.9 (Two forbidden exponential rays)**.** *Let $x \in \mathcal{S} \setminus \mathbb{Z}$, let $\vartheta = \log_2 3$, and let $0 \neq \gamma \in \overline{\mathbb{Q}}$. Then both*

$$e^{\gamma x} \qquad \text{and} \qquad e^{\gamma \vartheta}$$

are transcendental. [classical]

Proof. For the first value, apply five exponentials to the independent pairs $(1, x)$ and $(\log 2, \log 3)$. The four rectangular exponentials are $2, 3, M, A$, all algebraic, so the fifth $e^{\gamma x}$ is transcendental.

For the second value, use the independent pairs $(\log 2, \log 3)$ and $(1, x)$. The same four rectangular exponentials occur, and the fifth is $e^{\gamma \log 3 / \log 2} = e^{\gamma \vartheta}$. □

This does not prove the conjecture: γ must be algebraic, whereas the useful coefficients arising from new integer bases are themselves logarithms. It does show that a counterexample forces two entire algebraic rays of exponential values to be transcendental.

6 Rational and integral outputs inside the algebraic spectrum

The current Lean snapshot settles algebraicity throughout Γ (and excludes it outside Γ). A finer arithmetic question remains: if $\alpha = 2^r 3^s \in \Gamma$, when is α^x rational or an integer? For a fixed solution the answer is completely encoded by prime valuations, and the resulting Smith lattice is a strict refinement of the divisible-hull theorem.

6.1 The valuation matrix

Let $x \in \mathcal{S} \setminus \mathbb{Z}$ and write

$$M = 2^x, \qquad A = 3^x.$$

Let P be the finite union of the prime supports of M and A, and define column vectors

$$\mathbf{m} = (v_p(M))_{p\in P}, \qquad \mathbf{a} = (v_p(A))_{p\in P}.$$

Let

$$V_x = \begin{pmatrix}\mathbf{m} & \mathbf{a}\end{pmatrix} \in M_{P\times 2}(\mathbb{Z}).$$

For $q = (r,s)^T \in \mathbb{Q}^2$ put

$$\alpha(q) = 2^r 3^s.$$

Then the unique positive rational-power interpretation gives

$$\alpha(q)^x = M^r A^s = \prod_{p\in P} p^{r v_p(M) + s v_p(A)}. \tag{6.1}$$

Lemma 6.1 (Rank two)**.** *The two columns of V_x are linearly independent over $\mathbb{Q}$.*

Proof. If $r\mathbf{m} + s\mathbf{a} = 0$, then $M^r A^s = 1$. Taking real logarithms gives

$$x(r\log 2 + s\log 3) = 0.$$

Since $x > 0$ and $\log 3/\log 2$ is irrational, $r = s = 0$. □

Theorem 6.2 (Exact rational and integral auxiliary-output spectrum)**.** *Let x, M, A, V_x be as above and let $q = (r,s)^T \in \mathbb{Q}^2$. Then*

$$\alpha(q)^x \in \mathbb{Q}_{>0} \iff V_x q \in \mathbb{Z}^P,$$
$$\alpha(q)^x \in \mathbb{N} \iff V_x q \in \mathbb{N}_0^P.$$

Together with theorem 5.2, this classifies algebraic, rational, and integral outputs for every positive algebraic auxiliary base. ***[new paper proof]***

Proof. Equation (6.1) gives a finite product of primes with rational exponents. Choose a common denominator d. If this product is rational, raising to the d-th power and comparing ordinary rational prime valuations shows that every exponent in (6.1) is an integer. Conversely integer exponents give a rational number. Such a rational number is a positive integer exactly when all its prime valuations are nonnegative. □

Thus three nested spectra occur:

$$\begin{aligned} \mathcal{B}_{\mathrm{alg}}(x) &= \{\alpha(q) : q \in \mathbb{Q}^2\} = \Gamma, \\ \mathcal{B}_{\mathrm{rat}}(x) &= \{\alpha(q) : q \in \Lambda_x\}, \\ \mathcal{B}_{\mathrm{int}}(x) &= \{\alpha(q) : q \in \Lambda_x^+\}, \end{aligned}$$

where

$$\Lambda_x = \{q \in \mathbb{Q}^2 : V_x q \in \mathbb{Z}^P\}, \qquad \Lambda_x^+ = \{q \in \Lambda_x : V_x q \geq 0\}.$$

6.2 Smith normal form and the finite rational-output defect

Because V_x has rank two, Λ_x is a rank-two lattice in $\mathbb{Q}^2$ containing $\mathbb{Z}^2$ with finite quotient. Let

$$\delta_1(V_x) = \gcd\{\text{all entries of } V_x\},$$

and let

$$\delta_2(V_x) = \gcd\left\{\left|\det\begin{pmatrix} v_p(M) & v_p(A) \\ v_q(M) & v_q(A)\end{pmatrix}\right| : p, q \in P\right\}.$$

The second gcd is positive by theorem 6.1.

Theorem 6.3 (Smith classification)**.** *Let d_1, d_2 be the Smith invariant factors of V_x, so $d_1 = \delta_1(V_x)$, $d_1 \mid d_2$, and $d_1 d_2 = \delta_2(V_x)$. After a unimodular change of coordinates in $\mathbb{Q}^2$,*

$$\Lambda_x = \frac{1}{d_1}\mathbb{Z} \oplus \frac{1}{d_2}\mathbb{Z}.$$

Consequently

$$\Lambda_x/\mathbb{Z}^2 \cong \mathbb{Z}/d_1\mathbb{Z} \oplus \mathbb{Z}/d_2\mathbb{Z}, \qquad |\Lambda_x/\mathbb{Z}^2| = \delta_2(V_x).$$

[new paper proof]

Proof. Choose unimodular matrices U, W with

$$UV_xW = \begin{pmatrix} d_1 & 0 \\ 0 & d_2 \\ 0 & 0 \\ \vdots & \vdots \end{pmatrix}.$$

Since U preserves the integer lattice in the target and W preserves $\mathbb{Z}^2$ in the domain, the condition $V_x q \in \mathbb{Z}^P$ becomes $d_1 q_1', d_2 q_2' \in \mathbb{Z}$ in the transformed coordinates. The quotient and determinant-divisor formulae are the standard Smith-normal-form identities. □

The first invariant has a direct output interpretation: d_1 is the largest integer d for which both M and A are perfect d-th powers. For the least conditional solution β, affine primitivity gives $d_1 = 1$. In that case

$$\Lambda_\beta/\mathbb{Z}^2 \cong \mathbb{Z}/\delta_2(V_\beta)\mathbb{Z}.$$

Thus one integer, the gcd of all 2×2 valuation minors, measures the entire finite defect between rational β-outputs at algebraic $2,3$-radical bases and the obvious rational $2,3$-unit bases.

The integral cone Λ_x^+ is the intersection of this lattice with the rational polyhedral cone $V_x q \geq 0$. By Gordan's lemma it is a finitely generated normal affine semigroup. This is a useful exact replacement for trying to enumerate algebraic auxiliary bases one at a time.

6.3 Two translated solutions remove the fractional defect

The algebraic spectrum does not depend on x, but the rational-output lattice Λ_x can. Translation by one gives a sharp intersection identity.

Proposition 6.4 (Translation intersection)**.** *Let $x \in \mathcal{S} \setminus \mathbb{Z}$. Then*

$$\Lambda_x \cap \Lambda_{x+1} = \mathbb{Z}^2.$$

Equivalently, for $\alpha \in \overline{\mathbb{Q}}_{>0}$, if both α^x and α^{x+1} are rational, then

$$\alpha = 2^u 3^v \qquad (u, v \in \mathbb{Z}).$$

[new paper proof]

Proof. By the algebraic spectrum, write $\alpha = 2^r 3^s$. The quotient

$$\alpha = \frac{\alpha^{x+1}}{\alpha^x}$$

is rational. A rational number of the form $2^r 3^s$ with $r, s \in \mathbb{Q}$ must have $r, s \in \mathbb{Z}$, by ordinary prime valuations. The converse is immediate. ☐

Corollary 6.5 (Universal rational and integral spectra under failure)**.** *Assume the conjecture fails and let $\alpha \in \overline{\mathbb{Q}}_{>0}$.*

1. *α^y is rational for every nonintegral solution y if and only if $\alpha = 2^u 3^v$ with $u, v \in \mathbb{Z}$.*
2. *α^y is an integer for every solution y if and only if $\alpha = 2^u 3^v$ with $u, v \in \mathbb{N}_0$.*

Proof. For the first part, use any nonintegral x and $x + 1$ in theorem 6.4. Conversely rational $2, 3$-units have rational outputs $M^u A^v$.

For the second part, the forward implication first gives $\alpha = a/b \in \mathbb{Q}_{>0}$ in lowest terms. If $z = \alpha^x \in \mathbb{N}$ and $z(a/b)^n = \alpha^{x+n}$ is an integer for every $n \geq 0$, then b^n divides the fixed integer z for every n, so $b = 1$. An integer in Γ is exactly $2^u 3^v$ with $u, v \in \mathbb{N}_0$. The converse is immediate. ☐

7 The symmetric tower constants

The constants

$$\vartheta = \frac{\log 3}{\log 2}, \qquad \eta = \frac{1}{\vartheta} = \frac{\log 2}{\log 3}$$

generate two symmetric "next outputs"

$$E_3 = 3^{\vartheta}, \qquad E_2 = 2^{\eta}.$$

Neither individual constant is known to be transcendental. Six exponentials does, however, control the pair, and a hypothetical counterexample makes their arithmetic significance exact.

7.1 An unconditional strong dichotomy

Theorem 7.1 (At least one symmetric tower constant is strongly transcendental). *At least one of*

$$\vartheta \log 3 = \log E_3, \qquad \eta \log 2 = \log E_2$$

lies outside $\widetilde{\mathcal{L}}$. In particular, at least one of E_3, E_2 is transcendental. ***[new paper proof]***

Proof. The three numbers $1, \vartheta, \eta$ are linearly independent over $\overline{\mathbb{Q}}$. A relation $a + b\vartheta + c/\vartheta = 0$ would give $b\vartheta^2 + a\vartheta + c = 0$ with algebraic coefficients, impossible because ϑ is transcendental. The logarithms $\log 2, \log 3$ are linearly independent over $\overline{\mathbb{Q}}$ by Baker's theorem.

Apply Roy's strong six exponentials theorem to those three rows and two columns. Four of the six products are already logarithms of algebraic numbers:

$$\log 2, \quad \log 3, \quad \vartheta \log 2 = \log 3, \quad \eta \log 3 = \log 2.$$

Therefore one of the two remaining products $\vartheta \log 3$ and $\eta \log 2$ lies outside $\widetilde{\mathcal{L}}$. □

The ordinary six exponentials theorem gives the weaker transcendence dichotomy directly. The strong form is useful because it persists in the branch classification below.

7.2 Exact conditional classification of E_3

Let $x \in \mathcal{S} \setminus \mathbb{Z}$, $M = 2^x$, and $A = 3^x$.

Theorem 7.2 (The E_3 equivalences). *The following are equivalent:*

1. $E_3 = 3^\vartheta$ *is algebraic;*
2. $\vartheta \log 3 \in \widetilde{\mathcal{L}}$*;*
3. M *is* $\{2,3\}$*-smooth;*
4. *there exist* $a, b \in \mathbb{N}_0$ *with* $b > 0$ *such that*

$$M = 2^a 3^b, \qquad x = a + b\vartheta;$$

5. $x \in \mathbb{Q} + \mathbb{Q}\vartheta$.

If these conditions fail, then $\vartheta \log 3 \notin \widetilde{\mathcal{L}}$ and E_3 is log-strongly transcendental. If they hold, then

$$E_3^b = \frac{A}{3^a} \in \mathbb{Q}_{>0},$$

so E_3 is an explicit algebraic radical. ***[new paper proof]***

Proof. The implications (4)⇒(3)⇒(5) are immediate, since taking logarithms of $M = 2^a 3^b$ gives $x = a + b\vartheta$. Conversely, if (5) holds, write $x = u + v\vartheta$ with $u, v \in \mathbb{Q}$ and clear a common positive denominator d:

$$M^d = 2^p 3^q \qquad (p, q \in \mathbb{Z}).$$

Because the left side is a positive integer, prime valuations force $p, q \geq 0$ and $d \mid p, q$. Thus $u = a$ and $v = b$ for $a, b \in \mathbb{N}_0$; nonintegrality of x gives $b > 0$. Hence (5)⇒(4).

Suppose (1) holds. Apply six exponentials to rows $1, x, \vartheta$ and columns $\log 2, \log 3$. The six exponentials are

$$2,\ 3,\ M,\ A,\ 3,\ E_3,$$

all algebraic. Hence the three rows are rationally dependent, so $x = u + v\vartheta$ for $u, v \in \mathbb{Q}$. The implication (5)⇒(4) just proved now gives $M = 2^a 3^b$ with $a, b \in \mathbb{N}_0$ and $b > 0$. This proves (1)⇒(4), while (4) gives $A = 3^a E_3^b$ and hence (1).

It remains to prove the strong statement. If M is not smooth, then $\log 2, \log M, \log 3$ are linearly independent over $\mathbb{Q}$: a rational relation with a nonzero $\log M$ coefficient would put M in Γ, and an integer in Γ is smooth. Baker upgrades this to $\overline{\mathbb{Q}}$-linear independence. Therefore $1, x, \vartheta$ are $\overline{\mathbb{Q}}$-linearly independent after multiplication by $\log 2$. Roy's theorem applies to rows $1, x, \vartheta$ and columns $\log 2, \log 3$. The other five products lie in $\widetilde{\mathcal{L}}$, forcing $\vartheta \log 3 \notin \widetilde{\mathcal{L}}$. Thus failure of (3) implies failure of (2), so (2) implies (3); conversely, (1) plainly implies (2). □

7.3 The symmetric E_2 classification

Theorem 7.3 (The E_2 equivalences)**.** *The following are equivalent:*

1. $E_2 = 2^\eta$ *is algebraic;*
2. $\eta \log 2 \in \widetilde{\mathcal{L}}$*;*
3. A *is* $\{2,3\}$*-smooth;*
4. *there exist* $c, d \in \mathbb{N}_0$ *with* $c > 0$ *such that*

$$A = 2^c 3^d, \qquad x = d + c\eta;$$

5. $x \in \mathbb{Q} + \mathbb{Q}\eta$.

If these conditions fail, $\eta \log 2 \notin \widetilde{\mathcal{L}}$*. If they hold, then*

$$E_2^c = \frac{M}{2^d} \in \mathbb{Q}_{>0}.$$

[new paper proof]

Proof. Interchange 2 and 3, and interchange ϑ with η, in the proof of theorem 7.2. For the strong step, use the $\mathbb{Q}$-independence of $\log 3, \log A, \log 2$ when A is not smooth. □

Proposition 7.4 (The smooth cases are mutually exclusive)**.** *For a nonintegral solution,* M *and* A *cannot both be* $\{2,3\}$*-smooth. Consequently* E_3 *and* E_2 *cannot both be algebraic.*

Proof. Write

$$M = 2^a 3^b, \qquad A = 2^c 3^d.$$

Then

$$x = a + b\vartheta = d + c\eta.$$

Multiplying the second equality by ϑ and eliminating x gives

$$b\vartheta^2 + (a - d)\vartheta - c = 0.$$

The transcendence of ϑ forces $b = c = 0$ and $a = d$, hence $x = a \in \mathbb{Z}$, a contradiction. □

The current Lean development proves a more refined primitive-core exclusion by deriving a quadratic polynomial for ϑ. The argument above is its smooth-output shadow and will fit naturally into the kernel classification.

8 The solution-independent second-iterate kernel

The exact spectrum controls one application of the exponent x. The next question is which bases in Γ return to Γ after that application. This turns mixed second iterates into finite-dimensional linear algebra on prime valuations.

The current module `AlgebraicOutputLocus` already proves two substantial slices of this picture: every genuinely mixed $(2^i 3^j)^{x^2}$ with $i, j > 0$ is transcendental, and the algebraic rational-ratio outputs lie on one explicit external p-adic line (hence have lattice rank at most one). The kernel K below upgrades those nonnegative-lattice statements to an exact $\mathbb{Q}$-linear invariant, valid simultaneously for every nonintegral solution and every rational direction.

8.1 Definition and valuation description

For $x \in \mathcal{S} \setminus \mathbb{Z}$ define

$$K_x = \Big\{(r, s) \in \mathbb{Q}^2 : M^r A^s \in \Gamma\Big\}. \tag{8.1}$$

Equivalently, $q = (r, s)$ lies in K_x exactly when

$$\alpha(q)^x \in \Gamma, \qquad \alpha(q) = 2^r 3^s.$$

Because $q \mapsto \alpha(q)$ is injective, we will also identify K_x with the corresponding rank-at-most-one subgroup $\{\alpha(q) : q \in K_x\} \subseteq \Gamma$ when discussing power orbits. Let

$$\mathbf{m}_{\text{ext}} = (v_p(M))_{p \neq 2,3}, \qquad \mathbf{a}_{\text{ext}} = (v_p(A))_{p \neq 2,3}.$$

Only finitely many coordinates are nonzero.

Proposition 8.1 (Kernel formula and dimension)**.** *One has*

$$K_x = \ker\Big(\mathbb{Q}^2 \longrightarrow \mathbb{Q}^{(\mathbb{P}\setminus\{2,3\})},\ (r, s) \longmapsto r\mathbf{m}_{\text{ext}} + s\mathbf{a}_{\text{ext}}\Big). \tag{8.2}$$

Moreover:

1. $\dim_{\mathbb{Q}} K_x \le 1$;
2. K_x *contains no vector* (r, s) *with* $rs > 0$;
3. $K_x = \mathbb{Q}(1, 0)$ *exactly when* M *is smooth;*
4. $K_x = \mathbb{Q}(0, 1)$ *exactly when* A *is smooth.*

[new paper proof]

Proof. Membership of $M^r A^s$ in Γ is exactly the vanishing of every prime valuation outside $2, 3$, giving (8.2). At least one of the two external vectors is nonzero by theorem 7.4; hence the linear map has rank at least one and the kernel has dimension at most one.

If $r, s > 0$, any external prime occurring in M or A has positive valuation in $M^r A^s$; if $r, s < 0$, it has negative valuation. Thus no same-sign vector lies in the kernel. Finally, if M is smooth then the first external column is zero and the second is nonzero, so the kernel is the first coordinate axis; the converse follows by testing $(1, 0)$. The statement for A is symmetric. □

8.2 The kernel is independent of the nonintegral solution

The exact conditional monoid already suggests that the same obstruction should persist under translations and positive integer multiples. The algebraic spectrum gives a stronger and more conceptual result.

Theorem 8.2 (Global kernel invariance)**.** *Let $x, y \in \mathcal{S} \setminus \mathbb{Z}$. Then*

$$K_x = K_y.$$

For $q = (r, s) \in \mathbb{Q}^2$, the following are equivalent:

1. *$q \in K_x$;*
2. *$q \in K_y$;*
3. *$(2^r 3^s)^{xy}$ is algebraic.*

If these conditions fail, then

$$xy\,(r \log 2 + s \log 3) \notin \widetilde{\mathcal{L}}. \tag{8.3}$$

[new paper proof]

Proof. Put $\alpha = 2^r 3^s \in \Gamma$. Then $B = \alpha^x$ is algebraic. By the exact spectrum applied to the nonintegral solution y,

$$\alpha^{xy} = B^y \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad B \in \Gamma \quad \Longleftrightarrow \quad q \in K_x.$$

The left side is symmetric in x, y, so the same condition is equivalent to $q \in K_y$. If $B \notin \Gamma$, the strong spectrum applied to base B and exponent y gives

$$y \log B = xy \log \alpha \notin \widetilde{\mathcal{L}}.$$

□

Hence, under failure, there is a single global subspace K, not one kernel per solution. It measures exactly which rational directions in the $2, 3$-radical group survive one additional nonintegral exponentiation.

Corollary 8.3 (Strong mixed-base extension)**.** *Let x, y be nonintegral solutions and let $r, s \in \mathbb{Q}$ have the same nonzero sign. Then*

$$(2^r 3^s)^{xy}$$

is log-strongly transcendental. In particular 6^{xy} is log-strongly transcendental.

The Lean theorem `TwoBaseNonintegerSolution.transcendental_mixed_two_three_squared_exponents` already gives ordinary transcendence for positive integers r, s at $y = x$. The corollary above extends it to rational same-sign directions, to arbitrary pairs of nonintegral solutions x, y, and to the log-strong conclusion (8.3).

8.3 Möbius-orbit characterization

Theorem 8.4 (Möbius criterion)**.** *For $x \in \mathcal{S} \setminus \mathbb{Z}$, the following are equivalent:*

1. $K_x \neq \{0\}$;
2. *there exist* $u, v, r, s \in \mathbb{Q}$ *with*
$$us - vr \neq 0, \qquad x = \frac{u + v\vartheta}{r + s\vartheta}; \tag{8.4}$$
3. $x \in \mathrm{PGL}_2(\mathbb{Q}) \cdot \vartheta$;
4. *the four numbers* $1, \vartheta, x, x\vartheta$ *are linearly dependent over* $\mathbb{Q}$.

If these conditions hold, their $\mathbb{Q}$-span has dimension exactly three; otherwise it has dimension four. ***[new paper proof]***

Proof. If $(r, s) \in K_x \setminus \{0\}$, write
$$M^r A^s = 2^u 3^v$$
with $u, v \in \mathbb{Q}$. Taking logarithms and dividing by $\log 2$ gives
$$x(r + s\vartheta) = u + v\vartheta.$$
The denominator cannot vanish because ϑ is irrational. If $us - vr = 0$, the numerator and denominator are rationally proportional and x is rational, contrary to theorem 2.1. This proves (1)⇒(2), and (2)⇒(1) follows by exponentiating the same identity.

Equation (8.4) is precisely membership in the rational projective-linear orbit. After cross-multiplication it is a rational linear relation among $1, \vartheta, x, x\vartheta$. Conversely, any such relation must involve x or $x\vartheta$ because $1, \vartheta$ are independent, and rearrangement gives (8.4). Finally, two independent relations would make K_x two-dimensional, contradicting theorem 8.1; thus the dependent case has dimension exactly three. □

This is a genuinely different reformulation from the usual four-exponentials matrix. It asks whether the hypothetical exponent is a rational fractional-linear transform of the fixed transcendental number ϑ. The answer is not forced by the original hypotheses: the generic kernel branch has no such relation.

8.4 Four counterexample types

The external valuation vectors give an exhaustive four-way classification.

Theorem 8.5 (Kernel-type classification)**.** *Let $x \in \mathcal{S} \setminus \mathbb{Z}$. Exactly one of the following occurs.*

Type I. ***M smooth.*** *$M = 2^a 3^b$ with $b > 0$, $K = \mathbb{Q}(1, 0)$, $x = a + b\vartheta$, E_3 is algebraic, and E_2 is log-strongly transcendental.*

Type II. ***A smooth.*** *$A = 2^c 3^d$ with $c > 0$, $K = \mathbb{Q}(0, 1)$, $x = d + c\eta$, E_2 is algebraic, and E_3 is log-strongly transcendental.*

Type III. ***Common external valuation ray.*** *Both outputs are nonsmooth, but their external valuation vectors are proportional. There are coprime $d, e \in \mathbb{N}$ and a nonzero integer vector $\mathbf{c} \geq 0$ such that*
$$\mathbf{m}_{\mathrm{ext}} = d\mathbf{c}, \qquad \mathbf{a}_{\mathrm{ext}} = e\mathbf{c}.$$

Then

$$K = \mathbb{Q}(e, -d),$$

both E_3, E_2 are log-strongly transcendental, and for some $u, v \in \mathbb{Z}$,

$$\frac{M^e}{A^d} = 2^u 3^v, \qquad x = \frac{u + v\vartheta}{e - d\vartheta}. \tag{8.5}$$

Type IV. Independent external directions. *The two external valuation vectors are linearly independent. Then $K = \{0\}$, $x \notin \mathrm{PGL}_2(\mathbb{Q}) \cdot \vartheta$, the four numbers $1, \vartheta, x, x\vartheta$ are $\mathbb{Q}$-linearly independent, and both E_3, E_2 are log-strongly transcendental.*

[new paper proof]

Proof. If one external vector vanishes, theorems 7.2, 7.3 and 8.1 give Types I and II; they are mutually exclusive. If both are nonzero, they have rank one or two. In rank one, clear denominators and divide the two scale factors by their gcd to obtain the displayed coprime d, e and a common nonnegative integer vector $\mathbf{c}$. The kernel is then $\mathbb{Q}(e, -d)$. The external valuations cancel, so $M^e/A^d \in \Gamma \cap \mathbb{Q}_{>0}$. Its 2- and 3-valuations are integers, giving $u, v \in \mathbb{Z}$. Taking logarithms gives (8.5). Rank two gives zero kernel. The claims about E_3, E_2 follow because both outputs are nonsmooth. □

Type III says that outside the distinguished primes, M and A are powers of one common integer core:

$$M = 2^{a_2} 3^{a_3} C^d, \qquad A = 2^{b_2} 3^{b_3} C^e$$

with $a_2, a_3, b_2, b_3 \in \mathbb{N}_0$ and C supported only on primes outside $2, 3$. Type IV, by contrast, supplies two external primes p, q with nonzero valuation determinant

$$\Delta_{p,q} = v_p(M)\, v_q(A) - v_q(M)\, v_p(A) \neq 0. \tag{8.6}$$

This distinction is central to the proposed local determinant program.

9 Depth-three tower rigidity

The kernel gives a complete answer to the following question: starting from a positive algebraic base α, how many times can one repeatedly apply the same nonintegral exponent x before transcendence is forced?

For positive bases there is no ambiguity between iterated powers and powers of powers:

$$\left((\alpha^x)^x\right)^x = \alpha^{x^3}.$$

9.1 No nontrivial two-step return inside Γ

For $\alpha = 2^r 3^s \in \Gamma$ define $T_x(\alpha) = \alpha^x$. The map takes Γ into $\overline{\mathbb{Q}}_{>0}$, and K_x is exactly the set of elements whose image returns to Γ.

Lemma 9.1 (No two-step internal orbit)**.** *Let $x \in \mathcal{S} \setminus \mathbb{Z}$ and let $\alpha \in \Gamma \setminus \{1\}$. If $\alpha \in K_x$, then*

$$T_x(\alpha) \notin K_x.$$

Equivalently,

$$\alpha^x \in \Gamma \quad \Longrightarrow \quad \alpha^{x^2} \notin \Gamma.$$

[new paper proof]

Proof. Suppose both α and $\beta = T_x(\alpha)$ lie in the one-dimensional space $K_x \setminus \{0\}$. Under the identification $\Gamma \cong \mathbb{Q}^2$, two nonzero elements on the same rational line differ by a rational scalar, so $\beta = \alpha^q$ for some $q \in \mathbb{Q}$. But $\beta = \alpha^x$ and $\alpha \neq 1$, hence $x = q$, contradicting the irrationality of x. Thus $\beta \notin K_x$, which means $T_x(\beta) = \alpha^{x^2} \notin \Gamma$. □

The lemma has a useful dynamical interpretation. A nonzero direction in K may survive one return to the algebraic spectrum, but the returned direction cannot itself survive. There are no nontrivial cycles, fixed rays, or length-two paths entirely inside Γ.

9.2 Exact survival depth

Theorem 9.2 (Depth-three algebraic-tower theorem)**.** *Let $x \in \mathcal{S} \setminus \mathbb{Z}$ and let $\alpha \in \overline{\mathbb{Q}}_{>0} \setminus \{1\}$. At least one of*

$$\alpha^x, \qquad \alpha^{x^2}, \qquad \alpha^{x^3}$$

is transcendental. More precisely, the first transcendental term occurs at exactly one of the following depths:

1. ***Depth 1:*** *if $\alpha \notin \Gamma$, then α^x is log-strongly transcendental.*
2. ***Depth 2:*** *if $\alpha \in \Gamma$ but $\alpha \notin K_x$, then $\alpha^x \in \overline{\mathbb{Q}} \setminus \Gamma$ and α^{x^2} is log-strongly transcendental.*
3. ***Depth 3:*** *if $\alpha \in K_x \setminus \{1\}$, then $\alpha^x \in \Gamma$, $\alpha^{x^2} \in \overline{\mathbb{Q}} \setminus \Gamma$, and α^{x^3} is log-strongly transcendental.*

Equivalently, for some $j \in \{1, 2, 3\}$,

$$x^j \log \alpha \notin \widetilde{\mathcal{L}}.$$

[new paper proof]

Proof. Depth 1 is theorem 5.6. Suppose $\alpha \in \Gamma$. Then $\beta = \alpha^x$ is algebraic. If $\alpha \notin K_x$, then $\beta \notin \Gamma$; applying the strong spectrum to the algebraic base β and exponent x gives

$$\alpha^{x^2} = \beta^x$$

log-strongly transcendental.

If $\alpha \in K_x \setminus \{1\}$, then $\beta \in \Gamma$. Thus $\gamma = \beta^x = \alpha^{x^2}$ is algebraic. By theorem 9.1, $\beta \notin K_x$, hence $\gamma \notin \Gamma$. A final application of the strong spectrum to base γ gives $\gamma^x = \alpha^{x^3}$ log-strongly transcendental. □

This theorem is a compact synthesis of the auxiliary-base spectrum and the kernel. It also places a strict ceiling on how far an algebraic power tower can imitate the integral first level of a counterexample.

9.3 Exact behavior of bases 2, 3, and 6

Corollary 9.3 (Base 2 tower dichotomy). *For a nonintegral solution x with $M = 2^x$:*

1. *if M is not $\{2,3\}$-smooth, then 2^{x^2} is log-strongly transcendental;*
2. *if M is $\{2,3\}$-smooth, then 2^{x^2} is algebraic (indeed an integer) and 2^{x^3} is log-strongly transcendental.*

Proof. The vector $(1,0)$ lies in K_x exactly when $M \in \Gamma$, equivalently when the integer M is smooth. Apply theorem 9.2. In the smooth case, write $M = 2^a 3^b$; then

$$2^{x^2} = M^x = (2^x)^a (3^x)^b = M^a A^b \in \mathbb{N}.$$

□

The symmetric statement holds for base 3 and A.

Corollary 9.4 (Why base 6 always fails at depth two). *For every nonintegral solution x,*

$$x^2 \log 6 \notin \widetilde{\mathcal{L}},$$

so 6^{x^2} is log-strongly transcendental.

Proof. The coefficient vector $(1,1)$ has positive coordinate product and therefore does not lie in K_x. Since $6 \in \Gamma$, it has exact survival depth two. □

This recovers the current base-6 theorem and places it inside the kernel-verified family that every $(2^i 3^j)^{x^2}$ with $i, j > 0$ is transcendental. The new strengthening is the log-strong conclusion, its extension to arbitrary rational same-sign directions and pairs of nonintegral solutions, and the exact explanation of why the product base avoids the exceptional smooth axes.

9.4 New one-base tower equivalences of Alaoglu–Erdős

Theorem 9.5 (Universal three-level detector). *Fix any $\alpha \in \overline{\mathbb{Q}}_{>0} \setminus \{1\}$. The Alaoglu–Erdős conjecture is equivalent to*

$$\forall x \in \mathbb{R}, \quad 2^x, 3^x \in \mathbb{Z} \Longrightarrow \alpha^x, \alpha^{x^2}, \alpha^{x^3} \in \overline{\mathbb{Q}}. \tag{9.1}$$

The same equivalence holds with membership in $\widetilde{\mathcal{L}}$ required of the three logarithms $x^j \log \alpha$. ***[new paper proof]***

Proof. If the conjecture holds, every solution is a nonnegative integer, and all three powers are algebraic. If it fails, apply theorem 9.2 to a nonintegral solution. □

For bases in Γ, the first level is automatically algebraic. This gives a particularly simple new reformulation.

Corollary 9.6 (Base-2 cubic-tower equivalence). *The Alaoglu–Erdős conjecture is equivalent to*

$$\forall x \in \mathbb{R}, \quad 2^x, 3^x \in \mathbb{Z} \Longrightarrow 2^{x^2} \in \overline{\mathbb{Q}} \text{ and } 2^{x^3} \in \overline{\mathbb{Q}}. \tag{9.2}$$

The same statement holds with base 3 in place of base 2.

This differs from the repository's existing equivalent formulation requiring both 2^{x^2} and 3^{x^2} to be algebraic. A single original base suffices if one allows one more tower level.

9.5 A compact survival-depth table

For a nonintegral solution, the first forced transcendence depth of a positive algebraic base $\alpha \neq 1$ is:

Location of α	First log outside $\widetilde{\mathcal{L}}$	Structural reason
$\alpha \notin \Gamma$	$x \log \alpha$	six-exponential spectrum
$\alpha \in \Gamma \setminus K$	$x^2 \log \alpha$	first image exits Γ
$\alpha \in K \setminus \{1\}$	$x^3 \log \alpha$	no two-step internal orbit

This table is exact, not merely an upper bound.

10 A catalogue of equivalent formulations and near-equivalences

The corpus and the new results now provide a fairly rich dictionary. Let AE denote the original conjecture.

Formulation	Statement	Status
Original	$2^x, 3^x \in \mathbb{Z} \Rightarrow x \in \mathbb{Z}$.	Open
Candidate form	$M^\vartheta \in \mathbb{N} \Rightarrow M$ is a power of 2.	Exact
Localized radical form	No odd power-primitive $u > 1$ has $u^\vartheta \in \mathbb{Z}[1/3]$ with the canonical side conditions.	Lean/paper
No nontrivial generator	The conditional least generator β does not exist.	Lean
Base-5 algebraicity	Every solution has $5^x \in \overline{\mathbb{Q}}$.	Lean
Full algebraic output locus	For every positive algebraic γ, $\gamma^x \in \overline{\mathbb{Q}}$ iff $x \in \mathbb{Z}$ or $\gamma \in \Gamma$.	Lean
Arbitrary algebraic detector	For any fixed $\alpha \in \overline{\mathbb{Q}}_{>0} \setminus \Gamma$, every solution has $\alpha^x \in \overline{\mathbb{Q}}$.	New corollary of Lean locus
Both first iterates	Every solution has both $2^{x^2}, 3^{x^2} \in \overline{\mathbb{Q}}$.	Lean
Base-6 first iterate	Every solution has $6^{x^2} \in \overline{\mathbb{Q}}$.	Lean
Ratio-squared rank two	The algebraic locus among $(2^i/3^j)^{x^2}$ contains two noncollinear exponent vectors.	Lean
Single-base cubic tower	Every solution has $2^{x^2}, 2^{x^3} \in \overline{\mathbb{Q}}$.	New
Universal algebraic tower	For one fixed $\alpha \neq 1$, all three of $\alpha^x, \alpha^{x^2}, \alpha^{x^3}$ are algebraic for every solution.	New
Strong detector	For fixed $\alpha \notin \Gamma$, every solution has $x \log \alpha \in \widetilde{\mathcal{L}}$.	New
Subquadratic cumulative count	The candidate count is $o((\log X)^2)$ in the appropriate logarithmic scale, excluding a rank-two conditional monoid.	Paper equivalence
Sublinear block count	A sufficiently strong $o(\log X)$ dyadic candidate bound, with uniformity adequate to exclude the exact monoid.	Near-equivalence
Nonlinear closure	Closure of $\mathcal{S}$ under multiplication or squaring would imply AE, since nonintegral products are excluded under failure.	Paper implication

Formulation	Statement	Status
Four exponentials	The four exponentials conjecture applied to $(1, x)$ and $(\log 2, \log 3)$.	Classical implication

A few distinctions matter:

- The detector equivalences are truly pointwise: one fixed independent algebraic base is enough.
- The tower equivalence works even for bases inside Γ, where first-level algebraicity is automatic under a counterexample.
- A generic $O(\log X)$ block bound is not automatically contradictory; its leading constant and uniformity must beat the conditional monoid. An $o(\log X)$ bound is cleanly sufficient.
- Closure under a nonlinear operation is not known; the conditional theorem says exactly why obtaining it would be decisive.

11 Finite verification and computational audit

11.1 Formal and nonformal finite frontiers

Two finite statements must be kept distinct.

1. **[Lean]** The current formal development proves that no nonintegral solution occurs with $2^x < 4096 = 2^{12}$. Thus a counterexample has $x \geq 12$.
2. **[interval computation]** The third article [15] reports an exhaustive directed-rounding scan of every positive integer $M < 2^{26}$ and finds no candidate other than powers of 2. This gives the computational conclusion $x > 26$ for a counterexample.

The second result is stronger numerically but weaker epistemically. It should not be cited as a theorem of the Lean kernel.

11.2 What the MPFR scan must certify

For each M, one must decide whether

$$M^{\vartheta} = \exp\left(\frac{\log M \, \log 3}{\log 2}\right)$$

is an integer. Ordinary floating-point evaluation followed by a tolerance test is insufficient: a missed candidate could lie closer to an integer than the tolerance, and rounding error could cross the integer boundary.

The reported program instead constructs a rigorous interval enclosure using directed MPFR rounding. Conceptually, it computes lower and upper bounds for

$$y = \vartheta \log M, \qquad z = e^{y},$$

and verifies that the resulting interval contains no integer. A fast approximate locator selects nearby integer candidates, but the final accept/reject test uses the outward-rounded interval, not the approximate value. The code widens the locator neighborhood to protect against a one-unit mislocation.

11.3 Source-level audit findings

The source logic described in the third report has the right shape:

- lower and upper logarithms use opposite rounding directions;
- multiplication by the positive ratio $\log 3/\log 2$ preserves endpoint order;
- exponentiation is monotone and also outward rounded;
- every integer that could intersect the final interval is checked;
- exact powers of 2 are recognized as expected candidates rather than false failures;
- the loop bounds cover $1 \le M < 2^{26}$.

I found no conceptual rounding error in that design. However, a source audit is not an execution transcript, and the report's binary/hash record is not a compact independently checkable certificate of all 2^{26} interval exclusions.

11.4 How to turn the scan into a stronger artifact

A preferable next finite artifact would split the range into blocks and emit, for each block, a compact certificate containing:

- the MPFR precision and exact constants used;
- a hash of the source and compiler command;
- the minimum certified distance from an interval to the nearest integer;
- the input attaining that minimum;
- an independently replayable block checksum;
- optionally, rational interval endpoints for only the extremal near misses.

A checker substantially smaller than the search program could then replay the certificates. The ultimate formal route is a Lean interval proof for blocks using rational bounds for log and exp, but extending the theoretical structure is currently more valuable than pushing the raw cutoff by another few bits.

12 Lean formalization roadmap

The repository advanced during this audit. The current snapshot has already formalized the ordinary algebraic-output layer that an earlier draft of this report had listed as future work. The remaining roadmap therefore separates (i) newly available infrastructure, (ii) elementary algebraic consequences that can be added now, and (iii) genuinely stronger logarithmic transcendence statements requiring new classical inputs.

12.1 What the current snapshot just formalized

The new kernel-verified layer consists of:

- `AlgebraicSixExponentials`: a symmetric positive-real algebraic six-exponentials interface;
- `AlgebraicRationalBase` and `AlgebraicBaseUnitPower`: exact rational and arbitrary positive-algebraic output loci;
- `AlgebraicRankThreeEquivalence`: the rank-three algebraic-output reformulation of Alaoglu–Erdős;
- `AlgebraicOutputLocus`: mixed squared-output transcendence, the rank-one ratio locus, and an explicit external-prime valuation line;
- `CanonicalRadicalAlgebraicOutput`, `CanonicalRadicalDivisibleHull`, and `CanonicalRadicalFieldNorm`: corresponding upgrades for the canonical radicals.

Consequently the ordinary $2,3$-saturated algebraic spectrum is no longer a formalization proposal. The genuinely new formal targets in this report are the Smith refinement, the full rational kernel and its Möbius classification, the depth-three orbit theorem, and the log-strong upgrades.

12.2 What can plausibly be formalized now

12.2.1 A base-2 squared/cubed tower theorem

The base-2 specialization of theorem 9.3 only needs the current natural-base algebraicity classification, the smooth-output decomposition, and transcendence of ϑ. It does not require arbitrary algebraic bases or Roy's theorem. A suitable new module could be

`TwoBaseIntegerExponent/SquaredCubedTower.lean`.

The target interface is schematically:

```
/-- A nonintegral two-base solution cannot have algebraic
    base-2 outputs at both the squared and cubed exponents. -/
theorem TwoBaseNonintegerSolution.not_both_two_squared_cubed_algebraic
    {x : R} (hx : TwoBaseNonintegerSolution x) :
    not (IsAlgebraic Q ((2 : R) ^ (x * x)) and
         IsAlgebraic Q ((2 : R) ^ (x * x * x))) := by
  ...

/-- A one-base, two-level reformulation of Alaoglu--Erdos. -/
theorem alaogluErdosConjecture_iff_two_squared_cubed_algebraic :
    AlaogluErdosConjecture <->
      forall x : R,
        (exists z : Z, (z : R) = (2 : R) ^ x) ->
        (exists z : Z, (z : R) = (3 : R) ^ x) ->
        IsAlgebraic Q ((2 : R) ^ (x * x)) and
        IsAlgebraic Q ((2 : R) ^ (x * x * x)) := by
```

```
  ...
```

A direct proof splits on whether $M = 2^x$ is $\{2,3\}$-smooth. In the nonsmooth case, the current natural-base theorem makes $M^x = 2^{x^2}$ transcendental. In the smooth case, 2^{x^2} is an integer. If it were also smooth, expanding $x = a + b\vartheta$ and $(a + b\vartheta)^2$ would give a nonzero quadratic polynomial for ϑ; hence the integer base 2^{x^2} is nonsmooth, and its x-th power 2^{x^3} is transcendental.

12.2.2 External valuation rank and the four types

A module

`TwoBaseIntegerExponent/ExternalValuationKernel.lean`

should build directly on `AlgebraicOutputLocus`, which already extracts one external prime and proves a rank-one nonnegative-lattice slice. The new module can avoid quotient vector spaces initially. For natural outputs M, A, define the predicate that all external valuations of $M^r A^s$ cancel after clearing a common denominator. The rank-zero/one/two split can be expressed through proportionality of finitely supported valuation functions and the existence of a nonzero 2×2 minor. Useful target theorems are:

```
/-- Both nonsmooth outputs either share one external valuation ray
    or have a nonzero external two-prime minor. -/
theorem common_external_ray_or_exists_nonzero_minor ... :
  (exists d e C, ...) or
  (exists p q : Nat, p.Prime and q.Prime and
    p != 2 and p != 3 and q != 2 and q != 3 and
    padicValNat p M * padicValNat q A !=
      padicValNat q M * padicValNat p A) := by
  ...
```

Mathlib's finitely supported functions and unique factorization API should suffice. The main engineering issue is choosing a valuation representation that handles zero entries without repeated coercion friction.

12.2.3 Nonlinear closure from the third report

The polynomial identity argument is particularly formalization-friendly. Under failure, write every solution uniquely as $n + k\beta$. If

$$(n + k\beta)(m + \ell\beta) = r + s\beta,$$

then transcendence of β forces $k\ell = 0$. A module

`TwoBaseIntegerExponent/NonlinearClosure.lean`

could include:

- product closure iff one factor is integral;
- $P(\beta) \in \mathcal{S}$ iff P is affine with nonnegative integral coefficients at β;
- the rational-function field intersection;
- exact common-perfect-power degree $\gcd(n, k)$;
- primitive-pair counting after importing the necessary Möbius-sum estimates.

The first three need little analytic number theory and would materially tighten the formal conditional picture.

12.2.4 Smith-normal-form rational-output lattices

The valuation criterion theorem 6.2 is elementary once the algebraic base has already been represented as $2^r 3^s$. A first formal milestone can state it for rational pairs with an explicit common denominator, avoiding a general positive algebraic-number API. Mathlib already contains Smith normal form infrastructure in various matrix/module guises, but a bespoke rank-two theorem may be shorter and more stable.

12.3 What needs a larger transcendence formalization

12.3.1 General saturated-pair packaging

The specialized $2, 3$ statement is now kernel-verified. A lightweight optional module can package the same idea for arbitrary multiplicatively independent positive algebraic bases a, b:

$$\alpha^x \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \alpha \in \Gamma_{\mathrm{sat}}(a, b),$$

under $x \notin \mathbb{Q}$ and algebraicity of a^x, b^x. The existing generic theorem in `AlgebraicSixExponentials` supplies the difficult implication; the reverse implication is elementary rational-power algebra. This is primarily an API/generalization task, not a new determinant project.

12.3.2 The strong spectrum and symmetric constants

The assertions involving $\widetilde{\mathcal{L}}$ require formal versions of:

- Baker's homogeneous linear independence theorem for logarithms of algebraic numbers;
- Roy's rank theorem/strong six exponentials theorem;
- a coherent formal definition of the $\overline{\mathbb{Q}}$-linear logarithm span, including branches.

This is a substantial project in its own right. A pragmatic layering is to formalize the ordinary-transcendence versions first; every kernel and tower statement has a non-strong version using only the exact algebraic spectrum.

12.4 Suggested module order

A dependency-minimizing order is:

1. `NonlinearClosure.lean`;
2. `SquaredCubedTower.lean`;
3. `ExternalValuationKernel.lean`;
4. `AlgebraicRadicalOutputLattice.lean`;

5. `GeneralSaturatedBaseSpectrum.lean` (optional API packaging);
6. `StrongAlgebraicBaseSpectrum.lean`;
7. `SymmetricTowerConstants.lean`.

The first five reuse the current determinant and algebraic-output infrastructure. Only the last two require a formal Baker/Roy logarithm-span layer.

12.5 Proof-audit checklist for new formal modules

Each proposed theorem should explicitly track:

- positivity hypotheses required by real-power multiplication laws;
- whether "integer" is represented by $\mathbb{Z}$ or positive $\mathbb{N}$;
- the distinction between rational powers and natural powers;
- branch-free real logarithms versus complex logarithm spans;
- finite prime support before applying gcd or Smith arguments;
- the exact source of irrationality/transcendence of x and ϑ;
- status boundaries between kernel proof, imported classical theorem, and interval computation.

13 Branch-sensitive future research program

A single undifferentiated instruction to "use more transcendence theory" is not a research plan. The kernel classification suggests four different hypothetical worlds, with different arithmetic resources. The most promising next work should attack them separately and then recombine the results.

13.1 Priority 1: a two-prime tropical/p-adic determinant for Type IV

Type IV supplies external primes p, q with the nonzero valuation minor (8.6). This is information discarded by the current archimedean interpolation determinants.

For conditional lattice points (n_i, k_i) define synchronized integer outputs

$$m_i = 2^{n_i} M^{k_i}, \qquad a_i = 3^{n_i} A^{k_i}.$$

Choose column exponent pairs $(u_j, v_j) \in \mathbb{N}_0^2$ and form

$$D = \det\big(m_i^{u_j} a_i^{v_j}\big)_{0 \le i,j < r}. \tag{13.1}$$

At an external prime p,

$$v_p\Big(m_i^{u_j} a_i^{v_j}\Big) = k_i\big(u_j v_p(M) + v_j v_p(A)\big). \tag{13.2}$$

Thus every determinant term has a tropical assignment weight. Ordering the k_i and the column slopes

$$c_{p,j} = u_j v_p(M) + v_j v_p(A)$$

allows the rearrangement inequality to identify a minimal valuation permutation. A unique minimum, together with a nonzero residual determinant modulo p, gives an exact or sharp lower bound for $v_p(D)$. The independent q-slope ordering supplied by $\Delta_{p,q} \neq 0$ should provide a second nonarchimedean gain that cannot be absorbed into the first.

The target is an adelic product-formula contradiction:

$$|D|_\infty \prod_{\ell \in \{p,q\}} |D|_\ell < 1.$$

after all remaining norms are bounded by 1, while D is a nonzero integer. The archimedean factor comes from clustered nodes; the p- and q-adic factors come from the exact (n,k) lattice and external valuation directions.

Research target 13.1 (Tropical noncancellation lemma)**.** Choose a family of exponent pairs (u_j, v_j) for which the minimal assignment in (13.2) is unique at p and at q, or prove that the leading residual minor is a unit after a controlled column transformation.

Research target 13.2 (One-logarithm recovery)**.** Show that the combined nonarchimedean gain improves the unconditional dyadic candidate count from $O((\log X)^2)$ to $o(\log X)$ for configurations having a nonzero external valuation minor.

This is the highest-priority direction because Type IV is the generic branch and because it uses genuinely new arithmetic information rather than only enlarging an archimedean exponent box.

13.2 Priority 2: isolate and attack the common-core Type III branch

In Type III one has, after normalization,

$$M = 2^{a_2} 3^{a_3} C^d, \qquad A = 2^{b_2} 3^{b_3} C^e,$$

with one external primitive core $C > 1$ and coprime $d, e > 0$. The relation

$$M^e / A^d = 2^u 3^v$$

produces the Möbius identity (8.5).

This branch has no two-prime valuation area: every external valuation minor vanishes. Instead, it has an exact common-core quotient and a rational linear relation among $1, \vartheta, x, x\vartheta$.

Promising targets are:

1. derive a canonical normalization of C, d, e, a_i, b_i compatible with the repository's primitive output cores and prove uniqueness;
2. translate all perfect-power and distinguished-base depth inequalities into constraints on d, e, u, v in (8.5);
3. study whether the denominator $e - d\vartheta$ can be forced to give too good a rational approximation to ϑ once M, A are integers of the prescribed sizes;
4. build a one-core interpolation determinant over $\mathbb{Q}(C)$ whose conjugate or local norms retain information lost when all external vectors are proportional;

5. seek a reduction from Type III to one of the smooth axis types by proving that a power-primitive common core cannot occur on both output sides.

Research target 13.3 (Common-core exclusion)**.** Prove that a nonintegral two-base solution cannot have proportional nonzero external valuation vectors. Equivalently, eliminate Type III or force one output to be $\{2,3\}$-smooth.

Even a partial result such as excluding $d = e$, squarefree C, or a single external prime would materially narrow the conjecture.

13.3 Priority 3: pure-radical attacks on Types I and II

In Type I,

$$x = a + b\vartheta, \qquad E_3^b = A/3^a \in \mathbb{Q}_{>0},$$

so $E_3 = 3^\vartheta$ is an algebraic radical with explicitly bounded degree and known valuation data. Type II is symmetric for $E_2 = 2^{1/\vartheta}$.

Unlike the original relation, this is a legitimate number-field object. One can place E_3 in a pure extension, analyze all conjugates, and use norms. The current `AlgebraicIntegerDeterminant` and `AlgebraicOutputBridge` modules are directly relevant.

Concrete objectives are:

- prove that the exact degree of E_3 implied by the primitive normalization is incompatible with the 3-adic valuation of $A/3^a$;
- combine $2^\vartheta = 3$ with the minimal polynomial of E_3 in a two-variable interpolation determinant using both real embeddings and finite places;
- exploit the non-smoothness of A to obtain an external prime in the radical's norm;
- derive an algebraic auxiliary output outside Γ from a conjugate or norm of E_3; the spectrum theorem would then close the argument immediately.

The bare statement that E_3 should be transcendental is itself a difficult quadratic logarithm problem. The useful extra input is not merely algebraicity but the exact radical identity with an integer output and the canonical perfect-power constraints.

Research target 13.4 (Axis radical contradiction)**.** Assume $M = 2^a3^b$ with $b > 0$ and $A = 3^x \in \mathbb{N}$. Use the canonical primitive normalization to exclude the radical equality $(3^\vartheta)^b = A/3^a$.

13.4 Priority 4: an exact-lattice interpolation determinant

The current unconditional determinant estimates treat arbitrary clusters of candidates. A counterexample gives much more: every candidate has coordinates

$$m(n,k) = 2^n M^k, \qquad n,k \in \mathbb{N}_0.$$

The dyadic constraint is a thin strip

$$\log X \le n \log 2 + k \log M \le \log(2X)$$

through the integer lattice. Consecutive points have an ordered Sturmian geometry controlled by the continued fraction of $\beta = \log_2 M$.

Rather than selecting arbitrary nodes, construct the determinant directly from a complete strip segment and choose columns dual to its two lattice directions. Desired gains include:

- exact products of row gaps from the continued-fraction ordering;
- cancellation of the translation direction n before estimating the determinant;
- a determinant dimension proportional to the actual $O(\log X)$ strip count, not the square of an exponent-box side length;
- simultaneous use of output coordinates $3^n A^k$ and external prime valuations.

Research target 13.5 (Strip determinant)**.** Construct a nonzero integer determinant attached to all lattice points in one dyadic strip whose archimedean upper bound has an exponent linear, rather than quadratic, in the strip length.

This route may merge naturally with the Type-IV adelic determinant.

13.5 Priority 5: force an algebraic bridge outside Γ

The exact spectrum theorem turns the whole conjecture into a bridge-construction problem: under the original integer hypotheses, produce any positive algebraic $\alpha \notin \Gamma$ with algebraic α^x.

Potential bridge mechanisms include:

- a norm from an algebraic radical appearing in the canonical output normalization;
- an eigenvalue or determinant of a matrix built functorially from M, A whose x-th power is another algebraic matrix invariant;
- a resultant linking the two integer recurrences M^n and A^n;
- an algebraic limit forced by a sufficiently strong synchronized rational approximation;
- a local-to-global identity from two independent external primes.

The spectrum theorem also prevents wasted effort: a candidate bridge that simplifies to $2^r 3^s$ cannot solve the problem at first level.

13.6 Priority 6: denominator-sensitive orbit methods

Reducing kx modulo 1 gives the dense orbit

$$\left(2^{\{kx\}}, 3^{\{kx\}}\right) = \left(\frac{M^k}{2^{\lfloor kx \rfloor}}, \frac{A^k}{3^{\lfloor kx \rfloor}}\right).$$

The first coordinate is dyadic rational and the second triadic rational with synchronized exponent denominators. Generic rational-point counting on a transcendental curve is too weak: $O(\log H)$ points up to height H are compatible with known bounds.

A useful theorem must exploit the restricted denominator supports and their shared orbit index. Possible frameworks are:

- a quantitative S-integral Pila–Wilkie theorem with denominator support;
- a Subspace-Theorem estimate for local Taylor approximants to the curve $v = u^{\vartheta}$;
- an adelic determinant on the orbit points, retaining 2- and 3-adic denominators;
- a discrepancy estimate weighted by exact denominator growth rather than only orbit count.

13.7 Priority 7: computation as a certificate generator, not a substitute

Extending the scan from 2^{26} to 2^{30} or 2^{32} is straightforward engineering but unlikely to alter the theoretical bottleneck. More valuable computational work would:

- locate extremal near-candidates and test conjectured lower-bound shapes;
- search for the best determinant column sets by integer optimization;
- explore tropical valuation orderings for Type IV;
- enumerate possible small canonical normalizations (a, b, d, e) consistent with all formal inequalities;
- produce replayable interval certificates.

13.8 Recommended order of attack

The practical order I recommend is:

1. formalize the base-2 squared/cubed theorem and nonlinear closure package;
2. formalize the four kernel types at the natural-output valuation level;
3. prototype the Type-IV two-prime tropical determinant computationally;
4. isolate the exact Type-III common-core normal form and prove special-case exclusions;
5. revisit Types I/II with the existing algebraic-integer determinant machinery;
6. only then invest in a larger generic determinant or a deeper finite scan.

The branch split prevents a method suited to two independent external primes from being judged against a one-core exceptional case where it cannot possibly work.

14 Conclusion

The conjecture remains unproved in this analysis. The exact obstruction is visible and honest: the original four logarithms form the unresolved singular 2×2 configuration of four exponentials. The ProveIt corpus has nevertheless converted a hypothetical failure into an extraordinarily rigid object: a least transcendental generator, a free rank-two solution/output monoid, exact counting laws, radical normal forms, strong spacing bounds, and severe restrictions on all auxiliary outputs.

The main conceptual advance of the present report is that the auxiliary-output picture is now complete for *all positive algebraic bases*. Under a counterexample, algebraic output is possible exactly on the saturated $2, 3$-radical group Γ. Inside that group, prime valuations classify rational and integral output;

a global kernel of dimension at most one classifies algebraic second iterates; and no nontrivial direction survives twice. Hence every algebraic-base tower becomes transcendental by the third level, yielding new one-base reformulations of Alaoglu–Erdős.

The kernel also divides the hypothetical world into four arithmetically distinct branches. That is more than a taxonomy: it identifies the data each future method must use. The generic branch supplies a nonzero two-prime valuation area and invites an adelic tropical determinant. The common-core branch supplies a rational Möbius relation and a single primitive external core. The two axis branches turn one of the symmetric constants $3^{\log_2 3}$ and $2^{\log_3 2}$ into an explicit algebraic radical. Eliminating these branches separately is, in my assessment, the most promising path beyond the current one-logarithm determinant deficit.

A Classical transcendence inputs and their exact roles

This appendix states the external theorems used in the new paper proofs. It is included to make clear which conclusions are elementary consequences of the repository hypotheses and which depend on major classical transcendence results.

A.1 Gelfond–Schneider

If α is algebraic with $\alpha \neq 0, 1$ and z is algebraic irrational, then every value of α^z is transcendental. In this report it is used twice:

1. a nonintegral solution x is first shown irrational and then must be transcendental because 2^x is algebraic;
2. $\vartheta = \log_2 3$ is irrational and must be transcendental because $2^\vartheta = 3$.

The theorem says nothing directly when the exponent is already transcendental, which is why it cannot settle the original problem.

A.2 Baker's homogeneous theorem

A form sufficient here is: logarithms of algebraic numbers that are linearly independent over $\mathbb{Q}$ are linearly independent over $\overline{\mathbb{Q}}$. It is used to upgrade the row independence in theorem 5.6 and the logarithmic independence needed in theorems 7.2 and 7.3. The ordinary exact algebraic spectrum theorem 5.1 does not require this upgrade; rational independence suffices for six exponentials.

A.3 Six exponentials

If u_1, u_2, u_3 are linearly independent over $\mathbb{Q}$ and v_1, v_2 are linearly independent over $\mathbb{Q}$, then at least one of

$$e^{u_i v_j} \qquad (1 \leq i \leq 3,\ 1 \leq j \leq 2)$$

is transcendental. Its uses are:

- exact algebraic-base spectrum: rows $\log a, \log b, \log \alpha$, columns $1, x$;

- conditional algebraicity of E_3: rows $1, x, \vartheta$, columns $\log 2, \log 3$;
- the symmetric E_2 argument;
- the ordinary version of the unconditional E_3/E_2 dichotomy.

A.4 Five exponentials

For two independent pairs u_1, u_2 and v_1, v_2, and nonzero algebraic γ, at least one of

$$e^{u_1 v_1}, e^{u_1 v_2}, e^{u_2 v_1}, e^{u_2 v_2}, e^{\gamma u_2/u_1}$$

is transcendental. It supplies the forbidden rays in theorem 5.9. It does not remove the fifth exponential and hence does not prove four exponentials.

A.5 Roy's strong six exponentials theorem

Let $\widetilde{\mathcal{L}}$ be the $\overline{\mathbb{Q}}$-linear span of 1 and logarithms of algebraic numbers. If u_1, u_2, u_3 are independent over $\overline{\mathbb{Q}}$ and v_1, v_2 are independent over $\overline{\mathbb{Q}}$, then one product $u_i v_j$ lies outside $\widetilde{\mathcal{L}}$. This yields:

- the strong auxiliary-base spectrum;
- the strong symmetric-constant dichotomy;
- log-strong transcendence in the kernel and tower theorems.

Every ordinary transcendence conclusion in the report can be retained if this theorem is removed; only the strengthened "log outside $\widetilde{\mathcal{L}}$" clauses disappear.

A.6 Logical dependency graph

Result	Required input
Nonintegral x transcendental	Rational-integrality lemma + Gelfond–Schneider
Exact algebraic spectrum	Six exponentials
Strong spectrum	Exact row independence + Baker + Roy
Valuation lattice/Smith form	Exact spectrum + unique factorization
Kernel dimension and four types	Exact spectrum + elementary prime valuations
Kernel invariance	Exact spectrum applied twice
Möbius criterion	Kernel definition + real logarithms
Depth-three theorem	Exact spectrum + kernel dimension; Roy only for strong form
Base-2 tower equivalence	Natural-base algebraicity classification + smooth branch algebra
Symmetric constants	Six exponentials; Baker/Roy for strong form

B Further consequences of the kernel formalism

B.1 Orbit membership is global

Proposition B.1. *Assume the conjecture fails. Either every nonintegral solution belongs to* $\mathrm{PGL}_2(\mathbb{Q})\cdot\vartheta$*, or none does.*

Proof. By theorem 8.2, all nonintegral solutions have the same kernel. By theorem 8.4, Möbius-orbit membership is equivalent to nonzero kernel. □

This also follows from the exact solution monoid: if the least solution β lies in the orbit, then every $n + k\beta$ with $k > 0$ is a rational affine transform of an orbit point and therefore remains in the orbit. Kernel invariance is stronger because it does not need the least-generator theorem.

B.2 The kernel as the unique rational relation space

Define

$$R_x = \left\{(u, v, r, s) \in \mathbb{Q}^4 : u + v\vartheta - rx - sx\vartheta = 0\right\}.$$

Projection to (r, s) identifies R_x with K_x: the coefficients (u, v) are unique because $1, \vartheta$ are independent. Hence

$$\dim_{\mathbb{Q}} R_x = \dim_{\mathbb{Q}} K_x \in \{0, 1\}.$$

This gives a normalized linear-algebra version of the Möbius criterion. In Type IV the four normalized logarithmic products

$$1, \quad \vartheta, \quad x, \quad x\vartheta$$

are rationally independent even though the corresponding 2×2 real matrix has rank one. In Types I–III there is exactly one rational relation beyond the real determinant.

B.3 Exact delayed-survival subgroup

For a fixed nonintegral solution, define

$$\mathcal{D}_x = \{\alpha \in \Gamma : \alpha^{x^2} \in \overline{\mathbb{Q}}\}.$$

Then, under the coefficient identification $\Gamma \cong \mathbb{Q}^2$,

$$\mathcal{D}_x = K_x.$$

Thus:

- in Type IV, every nontrivial base in Γ fails by depth two;
- in Types I–III, exactly one rational rank-one subgroup survives to depth two;
- no nontrivial base survives to depth three while remaining algebraic at every prior level.

B.4 Mixed second iterates as a quotient-rank invariant

Let $\overline{\mathbb{Q}}_{>0}/\Gamma$ be viewed as a $\mathbb{Q}$-vector space. The two classes $[M], [A]$ span a subspace of dimension one or two. The kernel is the relation space of these classes:

$$K = \{(r, s) : r[M] + s[A] = 0\}.$$

Therefore

$$\dim K = 2 - \dim \operatorname{span}_{\mathbb{Q}}\{[M], [A]\}.$$

Type III and the two axes have quotient rank one; Type IV has quotient rank two. This coordinate-free viewpoint explains why the kernel is solution-independent: the algebraicity of α^{xy} detects the same quotient relation from either factor.

C Reproducibility inventory of the inspected corpus

C.1 Article sources

The three LaTeX blobs inspected at the repository commit on the title page were:

Path	Git blob SHA
`Article/alaoglu-erdos-combined-report/alaoglu-erdos-combined-report.tex`	`077723066e7a684a2c2cd1a7b0c49a02cfdf990b`
`Article/alaoglu-erdos-further-report/alaoglu-erdos-further-report.tex`	`d6bb12b447be426e675609b89a8b8b070e724b1e`
`Article/alaoglu-erdos-report-3/alaoglu-erdos-report-3.tex`	`048a3bcd902b5a09d06a68767e4aca7896aa0c33`

The reports themselves cite earlier base commits for some theorem inventories. This audit instead compared their claims with the current aggregate import at `4848c6d834190301e014028a9c78a3f0ef49b7b8`.

C.2 Current modules especially relevant to the new work

The current branch includes the following modules, among others:

- `ArithmeticRigidity.lean`;
- `AlgebraicConsequences.lean`;
- `AlgebraicIntegerDeterminant.lean`;
- `AlgebraicOutputBridge.lean`;
- `AlgebraicThirdBase.lean`;
- `AlgebraicSixExponentials.lean`;
- `AlgebraicRationalBase.lean`;

- `AlgebraicBaseUnitPower.lean`;
- `AlgebraicRankThreeEquivalence.lean`;
- `AlgebraicOutputLocus.lean`;
- `CanonicalRadicalAlgebraicOutput.lean`;
- `CanonicalRadicalDivisibleHull.lean`;
- `CanonicalRadicalFieldNorm.lean`;
- `CanonicalRadicalSharpness.lean`;
- `CanonicalRadicalThirdOutput.lean`;
- `CanonicalRadicalValuation.lean`;
- `CoreIndependence.lean`;
- `DenominatorGrowth.lean`;
- `FiniteAlgebraicOutputBridge.lean`;
- `RationalBaseThird.lean`;
- `RationalSixExponentials.lean`;
- `RationalThirdBase.lean`.

In particular, the current snapshot now proves the exact $2, 3$-divisible-hull locus for all positive algebraic auxiliary bases, together with rank restrictions on mixed second iterates. This report generalizes the ordinary locus to an arbitrary independent algebraic pair (a, b), refines rational and integral outputs by a Smith lattice, and strengthens the second-iterate slice to a solution-independent rational kernel and depth-three orbit theory.

C.3 Representative theorem-status map

Statement	Status	Location/type
Nonintegral solution is transcendental	Lean	core transcendence modules
No candidate below 2^{12}	Lean	finite exclusion modules
Exact conditional solution monoid	Lean	conditional structure modules
Exact finite trigonometric block averages	Lean	block Weyl modules
Continuous-test-function equidistribution	Paper	combined report
Arbitrary-node Vandermonde repulsion	Paper	further report
$O((\log X)^2)$ dyadic candidate bound	Paper	further report
Nonlinear product criterion in $\mathcal{S}$	Paper	third report
Primitive-solution quadratic asymptotic	Paper	third report
No candidate below 2^{26}	Computation	third-report MPFR source
Natural algebraic auxiliary-base classification	Lean	algebraic third-base modules
All positive algebraic auxiliary bases ($2, 3$ locus)	Lean	`AlgebraicBaseUnitPower`
General arbitrary-pair saturated spectrum	New paper	theorem 5.1

Statement	Status	Location/type
Strong $\widetilde{\mathcal{L}}$ spectrum	New paper	theorem 5.6
Valuation/Smith rational-output lattice	New paper	section 6
Mixed squared-output rank-one slices	Lean	`AlgebraicOutputLocus`
Global second-iterate kernel and four types	New paper	section 8
Depth-three tower theorem	New paper	section 9

D Common false shortcuts and audit warnings

The problem attracts several plausible but invalid one-line arguments. Recording them helps future formal and paper work avoid accidental circularity.

Caution D.1 (Fractional-part error). Writing $x = n + \alpha$ with $0 < \alpha < 1$ and $2^x \in \mathbb{Z}$ does *not* imply $2^\alpha \in \mathbb{Z}$. It implies only $2^\alpha \in \mathbb{Z}[1/2]$, and an irrational α is not an ordinary radical exponent $1/q$.

Caution D.2 (Rational-root circularity). From $M^\vartheta \in \mathbb{Z}$ one cannot conclude that ϑ is rational or that M is a perfect power. Gelfond–Schneider constrains algebraic irrational exponents, whereas ϑ is transcendental.

Caution D.3 (Baker is linear, not bilinear). The relation $\log M \log 3 = \log A \log 2$ is not a Baker linear form. Treating $\log 2$ and $\log 3$ as algebraic coefficients is invalid.

Caution D.4 (No modular real exponent). The expression $2^x \bmod p$ has no canonical meaning for a general real x. A p-adic exponential requires a p-adic exponent and a unit in its convergence domain; neither is supplied by the real equality.

Caution D.5 (Candidate quotients need not be candidates). If $m_1, m_2 \in \mathcal{C}$ and $m_1 \mid m_2$, it does not follow that $m_2/m_1 \in \mathcal{C}$, because the integer quotient of the outputs need not be integral. Multiplication is safe; division requires a separate valuation check.

Caution D.6 (Density alone is compatible). The orbit $\{kx\}$ is dense modulo one for irrational x, but the associated rational points have exponential height. An $O(\log H)$ sequence up to height H is compatible with generic transcendental-curve counting; density does not yield a contradiction without denominator sensitivity.

Caution D.7 (Numerical tolerance is not exclusion). Computing M^ϑ in floating point and checking its distance to the nearest integer against a fixed epsilon cannot certify a finite zero-free region. Outward interval bounds or an exact certificate are required.

Caution D.8 (Algebraicity of a power is branch-sensitive). From $z^d \in \overline{\mathbb{Q}}$ with integer $d > 0$ one may conclude $z \in \overline{\mathbb{Q}}$, but from $z^x \in \overline{\mathbb{Q}}$ with transcendental x no such algebraic closure argument is available.

E Notation index

Symbol	Meaning
$\mathcal{S}$	Real x with $2^x, 3^x \in \mathbb{Z}$.
$\mathcal{C}$	Positive integers M with $M^\vartheta \in \mathbb{N}$.
ϑ	$\log 3/\log 2 = \log_2 3$.
η	$1/\vartheta = \log_3 2$.

Symbol	Meaning
M, A	Integer outputs $M = 2^x$, $A = 3^x$.
β	Conditional least nonintegral solution.
$\overline{\mathbb{Q}}_{>0}$	Positive real algebraic numbers.
$\Gamma_{\mathrm{sat}}(a, b)$	$\{a^r b^s : r, s \in \mathbb{Q}\}$.
Γ	$\Gamma_{\mathrm{sat}}(2, 3)$.
$\widetilde{\mathcal{L}}$	$\overline{\mathbb{Q}}$-linear span of 1 and logarithms of algebraic numbers.
V_x	Full prime-valuation matrix with columns $\nu(M), \nu(A)$.
Λ_x	Rational coefficient lattice giving rational auxiliary outputs.
K_x	Kernel of external valuations, equivalently directions returning to Γ.
E_3, E_2	3^{ϑ} and $2^{1/\vartheta}$.

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